

TRIBHUVAN UNIVERSITY

INSTITUTE OF ENGINEERING (IOE)

ELECTRONIC DEVICES & CIRCUITS

Course Code: EX 151 / EX 501 / CT 501

BE Electronics, Information and Communication (BEI) & BE Computer (BCT)

Comprehensive Master Examination Solutions

Complete step-by-step mathematical derivations, small-signal AC models, DC biasing calculations, power amplifier efficiencies, oscillator analyses, and high-precision TikZ schematics for all exam sittings.

Academic Exam Papers Covered:

2083 Baishakh · 2082 Bhadra · 2082 Baishakh · 2081 Ashwin

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Chapter 1

2083 Baishakh Examination Solutions

Examination Overview

Level: Bachelor in Engineering (BE)

Full Marks: 60

Programme: BEI, BCT

Pass Marks: 24

Year / Part: I / II

Time: 3 Hours

Subject: Electronic Devices & Circuits (ENEX 151)

Examination Type: Back (New Course)

1.0.1 Question 1: BJT vs. FET Conduction & Common Collector Amplifier Design [1 + 5 = 6 Marks]

Problem Statement (2083 Baishakh, Q1)

Why BJT is bipolar and FET is unipolar? Design β -independent type DC biased common collector amplifier. Given parameters are: $V_{CC} = 20$ V, $I_C = 2$ mA, and $\beta = 100$. Use firm biasing method.

Part (a): Bipolar vs. Unipolar Operating Mechanism [1 Mark]

- **BJT is Bipolar:** Its current conduction relies simultaneously on **both majority and minority charge carriers** (free electrons and holes) flowing across two p - n junctions (emitter-base and collector-base).
- **FET is Unipolar:** Electric conduction takes place exclusively through a continuous channel containing **only one type of charge carrier** (electrons in n-channel, holes in p-channel) modulated by an electric field with zero minority carrier participation.

Part (b): β -Independent Common Collector (Emitter Follower) Design [5 Marks]

1. **Emitter Bias Voltage Selection (V_{EQ}):** To allow maximum symmetrical undistorted AC output

voltage swing without premature clipping:

$$V_{EQ} = \frac{V_{CC}}{2} = \frac{20 \text{ V}}{2} = \mathbf{10.0 \text{ V}}$$

2. **Emitter Resistor (R_E):** Since $I_E \approx I_C = 2.0 \text{ mA}$:

$$R_E = \frac{V_{EQ}}{I_E} = \frac{10.0 \text{ V}}{2.0 \text{ mA}} = 5.0 \text{ k}\Omega \quad (\text{Select standard } R_E = \mathbf{5.1 \text{ k}\Omega})$$

3. **Base DC Voltage (V_B):**

$$V_B = V_{EQ} + V_{BE} = 10.0 \text{ V} + 0.7 \text{ V} = \mathbf{10.7 \text{ V}}$$

4. **Firm Biasing Method ($I_{\text{bleed}} = 10I_B$):**

$$I_B = \frac{I_C}{\beta} = \frac{2.0 \text{ mA}}{100} = 0.02 \text{ mA} = 20 \mu\text{A}$$

$$I_{\text{bleed}} = 10 \times I_B = 10 \times 0.02 \text{ mA} = \mathbf{0.20 \text{ mA}}$$

5. **Voltage Divider Resistors (R_1, R_2):**

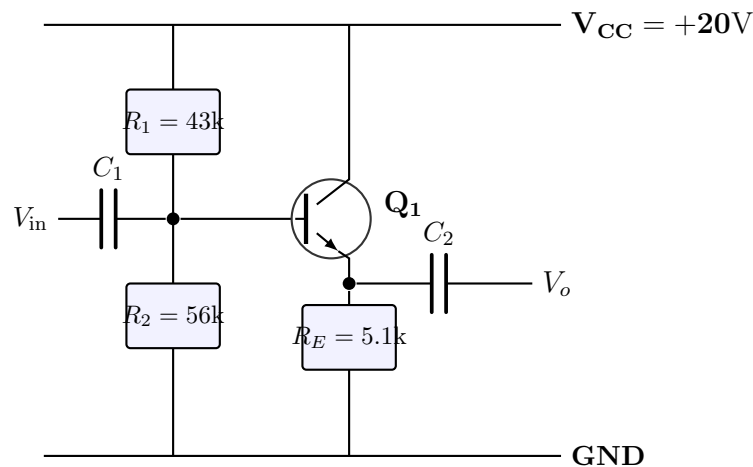
$$R_2 = \frac{V_B}{I_{\text{bleed}}} = \frac{10.7 \text{ V}}{0.20 \text{ mA}} = 53.5 \text{ k}\Omega \quad (\text{Select standard } R_2 = \mathbf{56 \text{ k}\Omega})$$

$$R_1 = \frac{V_{CC} - V_B}{I_{\text{bleed}} + I_B} = \frac{20 \text{ V} - 10.7 \text{ V}}{0.20 \text{ mA} + 0.02 \text{ mA}} = \frac{9.3 \text{ V}}{0.22 \text{ mA}} = 42.27 \text{ k}\Omega \quad (\text{Select standard } R_1 = \mathbf{43 \text{ k}\Omega})$$

6. **β -Independency Verification:**

$$R_{\text{th}} = R_1 \parallel R_2 = 43 \text{ k}\Omega \parallel 56 \text{ k}\Omega = \frac{43 \times 56}{43 + 56} = \mathbf{24.32 \text{ k}\Omega}$$

Firm bias stability criterion: $R_{\text{th}} \leq 0.1(1 + \beta)R_E = 0.1(101)(5.1 \text{ k}\Omega) = 51.5 \text{ k}\Omega \implies 24.32 \text{ k}\Omega < 51.5 \text{ k}\Omega \quad \checkmark$

Circuit Schematic / Analysis Diagram: Designed β -Independent Common Collector (Emitter Follower) Circuit**Final Verified Result**

$$R_1 = 43 \text{ k}\Omega, \quad R_2 = 56 \text{ k}\Omega, \quad R_E = 5.1 \text{ k}\Omega, \quad V_{EQ} = 10 \text{ V}, \quad I_{CQ} = 2.0 \text{ mA}$$

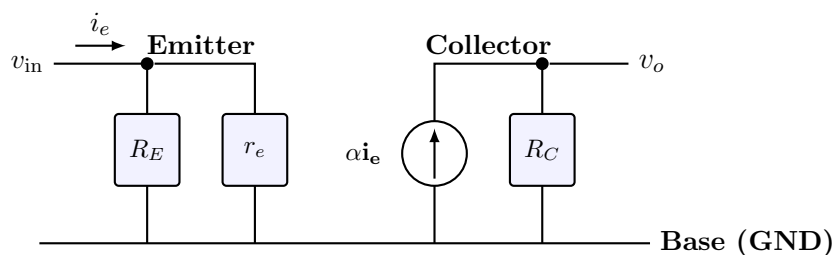
1.0.2 Question 2: Role of Bypass Capacitor & Common Base AC Parameters [2 + 5 = 7 Marks]**Problem Statement (2083 Baishakh, Q2)**

Explain the role of bypass capacitor in amplifier circuits. Draw the small signal equivalent circuit of a common base amplifier circuit and find its input impedance, output impedance and voltage gain.

Part (a): Role of Bypass Capacitor [2 Marks]

- In a BJT amplifier, the emitter resistor R_E provides crucial DC operating point stability against temperature variations and β changes.
- However, for AC signals, an unbypassed R_E introduces negative series current feedback which drastically reduces the AC voltage gain ($A_v \approx -R_C/R_E$).
- Connecting a **bypass capacitor (C_E)** in parallel with R_E provides a near zero-impedance AC short-circuit path ($X_{C_E} = \frac{1}{2\pi f C_E} \approx 0$) to ground at signal frequencies, eliminating AC negative feedback and maximizing voltage gain ($A_v \approx -R_C/r_e$).

Part (b): Common Base Small-Signal Equivalent Model [5 Marks]

Circuit Schematic / Analysis Diagram: Small-Signal r_e Model of Common Base (CB) Amplifier

1. **Input Impedance (Z_{in}):** Looking into the emitter terminal with base grounded:

$$Z_{in} = R_E \parallel r_e \approx r_e = \frac{V_T}{I_E} \quad (\text{very low, typically } 10 \, \Omega - 50 \, \Omega)$$

2. **Output Impedance (Z_{out}):** Looking back from the collector with $v_{in} = 0 \Rightarrow i_e = 0$:

$$Z_{out} = R_C \parallel r_o \approx R_C \quad (\text{high})$$

3. **Voltage Gain (A_v):**

$$v_o = -(-\alpha i_e)(R_C \parallel R_L) = \alpha i_e(R_C \parallel R_L), \quad v_{in} = i_e r_e$$

$$A_v = \frac{v_o}{v_{in}} = \frac{\alpha(R_C \parallel R_L)}{r_e} \approx +\frac{R_C \parallel R_L}{r_e} \quad (\text{Non-inverting, } 0^\circ \text{ phase shift})$$

4. **Current Gain (A_i):**

$$A_i = \frac{i_o}{i_e} = -\alpha \approx -1$$

Final Verified Result

$$Z_{in} \approx r_e = \frac{V_T}{I_E}, \quad Z_{out} \approx R_C, \quad A_v = +\frac{\alpha(R_C \parallel R_L)}{r_e}, \quad A_i = -\alpha \approx -1$$

1.0.3 Question 3: N-Channel Depletion-Type MOSFET [6 Marks]

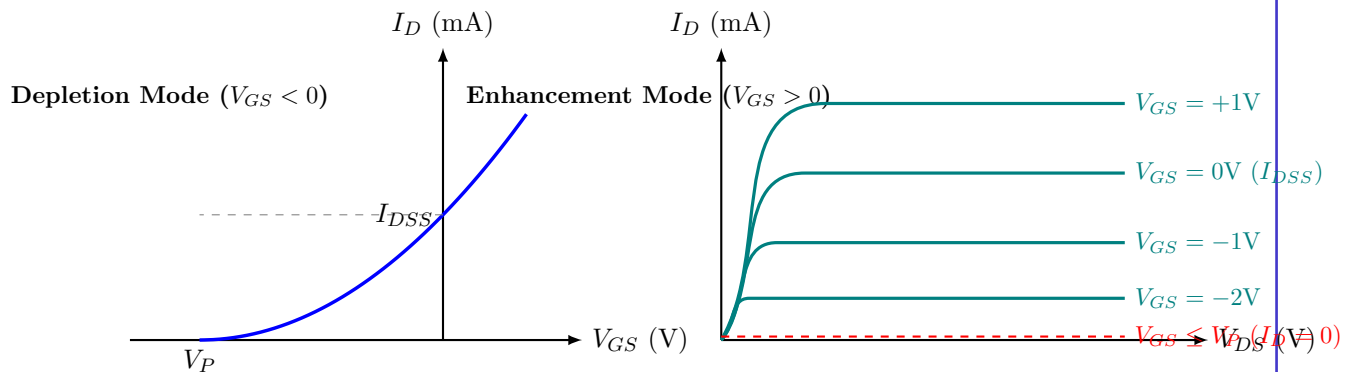
Problem Statement (2083 Baishakh, Q3)

Describe the construction and working principle of N Channel D-MOSFET with the help of characteristic curves and mathematical expressions.

1. **Physical Construction:** Fabricated on a p-type silicon substrate with two heavily doped n^+ Source and Drain regions, physically connected by an initial pre-diffused n-channel. The Gate electrode is completely insulated from the channel by an ultrathin silicon dioxide (SiO_2) dielectric layer ($R_{in} >$

$10^{12} \Omega$).

Circuit Schematic / Analysis Diagram: Transfer and Drain Characteristics of N-Channel D-MOSFET



2. Working Principle & Modes of Operation:

- Depletion Mode ($V_{GS} < 0$):** Negative gate voltage repels electrons out of the n-channel, creating a depletion region and reducing drain current below I_{DSS} . At $V_{GS} = V_P$ (pinch-off), the channel is completely closed ($I_D = 0$).
- Enhancement Mode ($V_{GS} > 0$):** Positive gate voltage attracts additional free electrons into the channel, widening it and increasing drain current above I_{DSS} .

3. Governing Shockley Equation:

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2 \quad \text{for } V_{DS} \geq V_{GS} - V_P$$

$$g_m = \left. \frac{\partial I_D}{\partial V_{GS}} \right|_{V_{DS}} = \frac{2I_{DSS}}{|V_P|} \left(1 - \frac{V_{GS}}{V_P} \right) = g_{m0} \left(1 - \frac{V_{GS}}{V_P} \right)$$

Final Verified Result

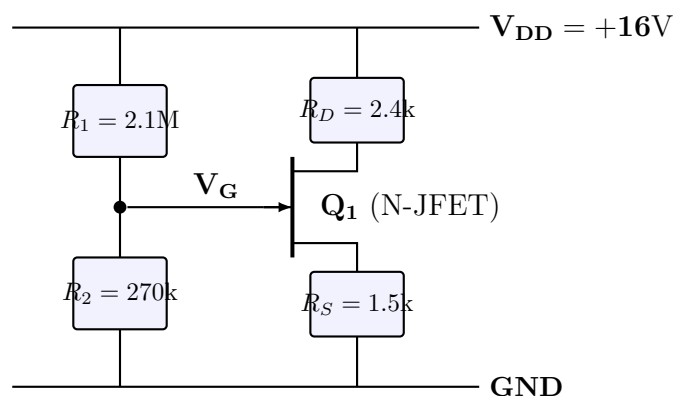
$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2 \quad \text{valid in both Depletion ($V_{GS} < 0$) and Enhancement ($V_{GS} > 0$) modes}$$

1.0.4 Question 4: JFET Voltage Divider Bias Analysis [7 Marks]

Problem Statement (2083 Baishakh, Q4)

Find I_D and V_{DS} for the given circuit. Given parameters: $V_P = -4$ V and $I_{DSS} = 8$ mA. Circuit values: $V_{DD} = +16$ V, $R_1 = 2.1$ M Ω , $R_2 = 270$ k Ω , $R_D = 2.4$ k Ω , $R_S = 1.5$ k Ω .

Circuit Schematic / Analysis Diagram: JFET Voltage-Divider Bias Circuit



1. Thevenin Gate DC Voltage (V_G): Since the gate of a JFET draws negligible current ($I_G \approx 0$):

$$V_G = V_{DD} \left(\frac{R_2}{R_1 + R_2} \right) = 16 \text{ V} \times \left(\frac{270 \text{ k}\Omega}{2100 \text{ k}\Omega + 270 \text{ k}\Omega} \right) = 16 \times \frac{270}{2370} = \mathbf{1.823 \text{ V}}$$

2. Gate-to-Source Voltage Equation (V_{GS}): Applying KVL around the gate-source loop:

$$V_{GS} = V_G - I_D R_S = 1.823 - 1.5 I_D \quad (\text{with } I_D \text{ in mA and } R_S = 1.5 \text{ k}\Omega)$$

3. Solving via Shockley's Equation:

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2 = 8 \left(1 - \frac{1.823 - 1.5 I_D}{-4} \right)^2 = 8 \left(\frac{4 + 1.823 - 1.5 I_D}{4} \right)^2$$

$$I_D = \frac{8}{16} (5.823 - 1.5 I_D)^2 = 0.5 (33.907 - 17.469 I_D + 2.25 I_D^2)$$

$$2 I_D = 33.907 - 17.469 I_D + 2.25 I_D^2 \implies \mathbf{2.25 I_D^2 - 19.469 I_D + 33.907 = 0}$$

Solving using the quadratic formula:

$$I_D = \frac{19.469 \pm \sqrt{(-19.469)^2 - 4(2.25)(33.907)}}{2(2.25)} = \frac{19.469 \pm \sqrt{379.04 - 305.16}}{4.5} = \frac{19.469 \pm 8.595}{4.5}$$

Two mathematical roots:

- $I_{D1} = \frac{28.064}{4.5} = 6.236 \text{ mA} \implies V_{GS1} = 1.823 - 1.5(6.236) = -7.53 \text{ V} < V_P = -4 \text{ V}$ (**Physically**

Invalid, beyond pinch-off).

- $I_{D2} = \frac{10.874}{4.5} = 2.416 \text{ mA} \implies V_{GS2} = 1.823 - 1.5(2.416) = -1.802 \text{ V}$ (Physically Valid, in active channel region).

4. Calculation of Drain-to-Source Voltage (V_{DS}):

$$V_{DS} = V_{DD} - I_D(R_D + R_S) = 16 \text{ V} - 2.416 \text{ mA} \times (2.4 \text{ k}\Omega + 1.5 \text{ k}\Omega) = 16 - 2.416(3.9) = 6.576 \text{ V}$$

Pinch-Off Verification:

$$V_{DS} = 6.576 \text{ V} \geq V_{GS} - V_P = -1.802 \text{ V} - (-4.0 \text{ V}) = 2.198 \text{ V} \implies \text{Confirmed in Saturation Region! } \checkmark$$

Final Verified Result

$$I_D = 2.416 \text{ mA}, \quad V_{GS} = -1.802 \text{ V}, \quad V_{DS} = 6.576 \text{ V}$$

1.0.5 Question 5: RC vs. LC Oscillators & Wien Bridge Oscillator [1 + 5 = 6 Marks]

Problem Statement (2083 Baishakh, Q5)

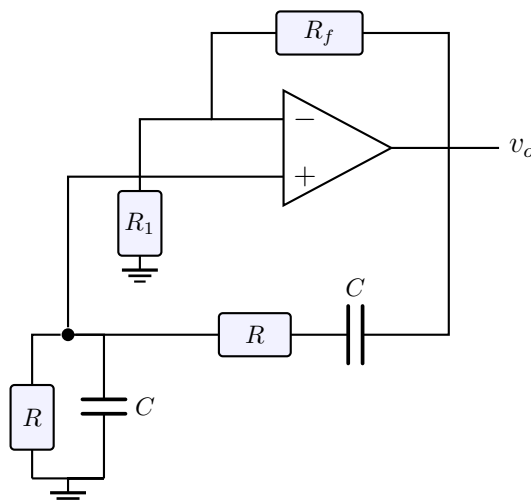
What is the main difference between RC and LC oscillator circuits in terms of application? Explain the operation of Wien Bridge Oscillator circuit with the help of circuit diagram and necessary mathematical expressions.

Part (a): Application Differences Between RC and LC Oscillators [1 Mark]

- **RC Oscillators (e.g., Wien Bridge, Phase-Shift):** Used primarily for ****Audio Frequencies (AF)**** (10 Hz – 1 MHz) where resistor-capacitor networks provide compact size, high low-frequency stability, and smooth sinusoidal waveforms.
- **LC Oscillators (e.g., Hartley, Colpitts):** Used primarily for ****Radio Frequencies (RF)**** (100 kHz – 500 MHz) where compact inductors and capacitors form high- Q resonant tank circuits. At audio frequencies, LC inductors become excessively bulky and lossy.

Part (b): Wien Bridge Oscillator Operation & Derivations [5 Marks]

Circuit Schematic / Analysis Diagram: Op-Amp Wien Bridge Oscillator Circuit with RC Lead-Lag Network



Derivation of Resonant Frequency and Gain Requirement: The transfer function of the lead-lag feedback network $\beta(s) = \frac{v_f(s)}{v_o(s)}$ is:

$$\beta(s) = \frac{Z_p}{Z_s + Z_p} = \frac{\frac{R}{1+sRC}}{\left(R + \frac{1}{sRC}\right) + \frac{R}{1+sRC}} = \frac{sRC}{(sRC)^2 + 3sRC + 1}$$

Substituting $s = j\omega$:

$$\beta(j\omega) = \frac{j\omega RC}{(1 - \omega^2 R^2 C^2) + 3j\omega RC}$$

1. **Oscillation Frequency (f_0):** For zero phase shift ($\angle\beta = 0^\circ$), the imaginary term in the bracket must vanish:

$$1 - \omega_0^2 R^2 C^2 = 0 \implies \omega_0 = \frac{1}{RC} \implies f_0 = \frac{1}{2\pi RC}$$

2. **Attenuation at Resonance:**

$$\beta(j\omega_0) = \frac{j}{3j} = \frac{1}{3}$$

3. **Barkhausen Gain Requirement ($|A_v\beta| \geq 1$):**

$$A_v = 1 + \frac{R_f}{R_1} \geq \frac{1}{\beta} = 3 \implies R_f \geq 2R_1$$

Final Verified Result

$$f_0 = \frac{1}{2\pi RC}, \quad R_f \geq 2R_1 \quad (|A_v| \geq 3)$$

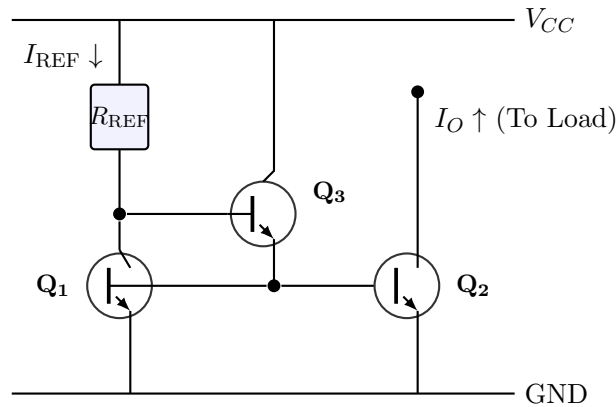
1.0.6 Question 6: Base-Current-Compensated Current Mirror [5 + 2 = 7 Marks]

Problem Statement (2083 Baishakh, Q6)

Explain how a base current compensation circuit reduces the difference between I_O and I_{REF} in a simple current mirror circuit.

1. Circuit Description & Role of Compensating Transistor Q_3 : In a standard two-transistor current mirror, the reference branch must supply the base currents of both Q_1 and Q_2 , introducing a relative error of $\approx 2/\beta$. A 3-transistor compensated current mirror inserts an emitter-follower transistor Q_3 to buffer the base currents.

Circuit Schematic / Analysis Diagram: Base-Current-Compensated BJT Current Mirror Circuit

2. Mathematical Derivation of Output Current I_O :

1. Assuming matched transistors ($I_{C1} = I_{C2} = I_O$ and $I_{B1} = I_{B2} = I_B = \frac{I_O}{\beta}$):

$$I_{E3} = I_{B1} + I_{B2} = 2I_B = \frac{2I_O}{\beta}$$

2. Transistor Q_3 base current is reduced by a factor of $(1 + \beta)$:

$$I_{B3} = \frac{I_{E3}}{1 + \beta} = \frac{2I_O}{\beta(1 + \beta)} \approx \frac{2I_O}{\beta^2}$$

3. Applying KCL at the reference collector node:

$$I_{REF} = I_{C1} + I_{B3} = I_O + \frac{2I_O}{\beta(\beta + 1)} = I_O \left[1 + \frac{2}{\beta^2 + \beta} \right]$$

$$I_O = \frac{I_{REF}}{1 + \frac{2}{\beta^2 + \beta}} \approx I_{REF} \left(1 - \frac{2}{\beta^2} \right)$$

3. Error Comparison:

- Simple 2-BJT Mirror Error: $\frac{|I_{REF}-I_O|}{I_{REF}} \approx \frac{2}{\beta} = 2.0\%$ (for $\beta = 100$).
- Base-Compensated Mirror Error: $\frac{|I_{REF}-I_O|}{I_{REF}} \approx \frac{2}{\beta^2} = 0.02\%$ (a **100-fold accuracy improvement**).

Final Verified Result

$$I_O = \frac{I_{REF}}{1 + \frac{2}{\beta^2 + \beta}} \approx I_{REF} \left(1 - \frac{2}{\beta^2}\right) \Rightarrow \text{Error reduced from } O(1/\beta) \text{ to } O(1/\beta^2)$$

1.0.7 Question 7: Class B Power Amplifier Calculations [6 Marks]**Problem Statement (2083 Baishakh, Q7)**

For a class B amplifier providing a 20 V peak signal to a 16 Ω load (speaker) and a power supply of $V_{CC} = 30$ V, determine the input power, output power, and circuit efficiency.

1. Peak Output Voltage (V_p):

$$V_p = 20.0 \text{ V}$$

2. AC Output Signal Power ($P_{o(ac)}$):

$$P_{o(ac)} = \frac{V_p^2}{2R_L} = \frac{(20 \text{ V})^2}{2 \times 16 \Omega} = \frac{400}{32} = 12.5 \text{ W}$$

3. Average DC Supply Current (I_{dc}): For a complementary Class B push-pull amplifier, each transistor conducts over half a cycle:

$$I_{dc} = \frac{2}{\pi} I_p = \frac{2}{\pi} \left(\frac{V_p}{R_L} \right) = \frac{2}{\pi} \left(\frac{20 \text{ V}}{16 \Omega} \right) = \frac{40}{16\pi} = \frac{2.5}{\pi} \approx 0.7958 \text{ A}$$

4. DC Input Power ($P_{i(dc)}$):

$$P_{i(dc)} = V_{CC} I_{dc} = 30 \text{ V} \times 0.7958 \text{ A} = \frac{75}{\pi} \approx 23.873 \text{ W}$$

5. Circuit Conversion Efficiency (η):

$$\eta = \frac{P_{o(ac)}}{P_{i(dc)}} \times 100\% = \frac{12.5 \text{ W}}{23.873 \text{ W}} \times 100\% = \frac{\pi}{4} \left(\frac{V_p}{V_{CC}} \right) \times 100\% = \frac{\pi}{4} \left(\frac{20 \text{ V}}{30 \text{ V}} \right) \times 100\% = 52.36\%$$

6. Total Transistor Power Dissipation ($P_{d(total)}$):

$$P_{d(total)} = P_{i(dc)} - P_{o(ac)} = 23.873 \text{ W} - 12.5 \text{ W} = 11.373 \text{ W} \quad (5.687 \text{ W per transistor})$$

Final Verified Result

$$P_{o(ac)} = 12.50 \text{ W}, \quad P_{i(dc)} = 23.87 \text{ W}, \quad \eta = 52.36\%$$

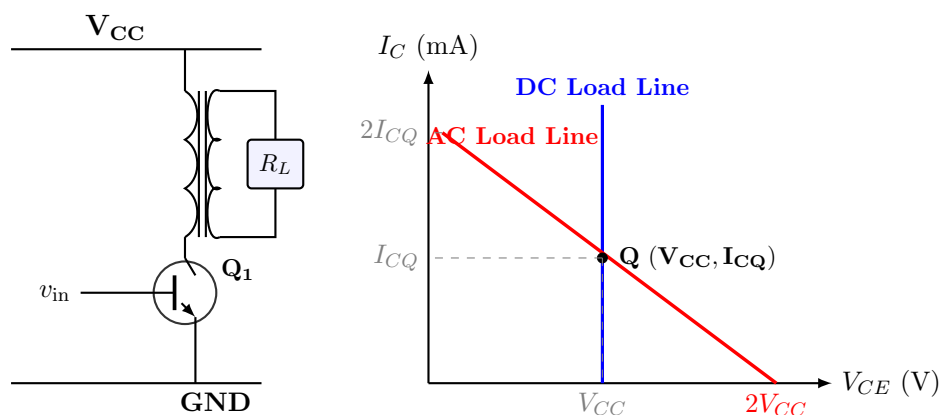
1.0.8 Question 8: Transformer-Coupled Class A Power Amplifier [7 Marks]

Problem Statement (2083 Baishakh, Q8)

Draw the circuit diagram and the characteristic curve of a transformer-coupled class A amplifier and derive its general and maximum efficiency.

1. Circuit Operation: A step-down output transformer matches the high collector output impedance to a low-impedance loudspeaker (R_L). The reflected AC load resistance seen by the primary is $R'_L = \left(\frac{N_1}{N_2}\right)^2 R_L$.

Circuit Schematic / Analysis Diagram: Transformer-Coupled Class A Amplifier Circuit & Dynamic AC/DC Load Line Curves



2. Efficiency Derivation:

1. **DC Input Power:** At quiescent bias $V_{CEQ} = V_{CC}$ (since primary DC winding resistance is $\approx 0 \Omega$):

$$P_{i(dc)} = V_{CC} I_{CQ}$$

2. **AC Output Power:**

$$P_{o(ac)} = \frac{V_p I_p}{2} = \frac{(V_{\max} - V_{\min})(I_{\max} - I_{\min})}{8}$$

3. General Efficiency (η):

$$\eta = \frac{P_{o(ac)}}{P_{i(dc)}} = \frac{\frac{V_p I_p}{2}}{V_{CC} I_{CQ}} = \frac{1}{2} \left(\frac{V_p}{V_{CC}} \right) \left(\frac{I_p}{I_{CQ}} \right) \times 100\%$$

4. **Maximum Efficiency ($\eta_{\max}$):** For maximum symmetrical undistorted output swing, $V_p = V_{CC}$ (voltage swings between 0 V and $2V_{CC}$) and $I_p = I_{CQ}$:

$$\eta_{\max} = \frac{1}{2}(1)(1) \times 100\% = 50.0\%$$

Final Verified Result

$$\eta = \frac{1}{2} \left(\frac{V_p}{V_{CC}} \right) \left(\frac{I_p}{I_{CQ}} \right) \times 100\%, \quad \eta_{\max} = 50.0\%$$

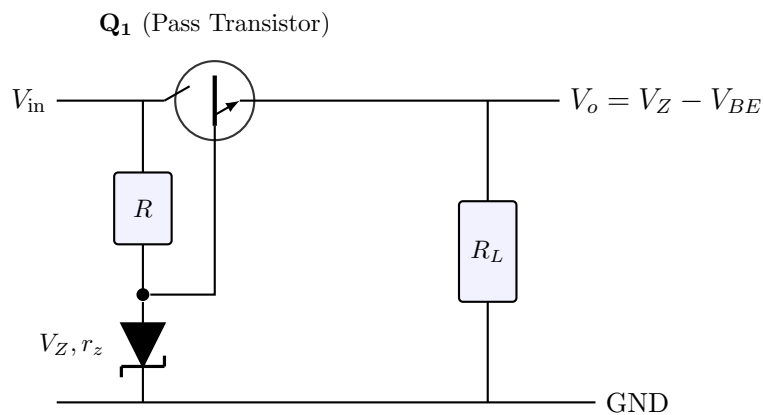
1.0.9 Question 9: Stability Factor of Discrete Series Voltage Regulator [4 Marks]

Problem Statement (2083 Baishakh, Q9)

Derive the expression for stability factor of a discrete-transistor series regulator.

1. **Circuit Model & Voltage Stability Factor Definition:** The Voltage Stability Factor (S_V) defines the line regulation capability of the series voltage regulator:

$$S_V = \left. \frac{\Delta V_o}{\Delta V_{in}} \right|_{R_L = \text{constant}}$$

Circuit Schematic / Analysis Diagram: Discrete Transistor Series Voltage Regulator Small-Signal Circuit

2. Derivation of Stability Factor S_V :

1. The Zener diode has dynamic internal resistance r_z . A change in input voltage ΔV_{in} produces a change in Zener voltage ΔV_Z through the voltage divider formed by R and r_z :

$$\Delta V_Z = \Delta V_{in} \left(\frac{r_z}{R + r_z} \right)$$

2. The output voltage follows the base voltage via the emitter-follower pass transistor Q_1 :

$$V_o = V_Z - V_{BE} \implies \Delta V_o = \Delta V_Z - \Delta V_{BE}$$

3. Since the base-emitter voltage drop $V_{BE} \approx 0.7$ V remains virtually constant ($\Delta V_{BE} \approx 0$):

$$\Delta V_o \approx \Delta V_Z = \Delta V_{in} \left(\frac{r_z}{R + r_z} \right)$$

4. Therefore, the voltage stability factor is:

$$S_V = \frac{\Delta V_o}{\Delta V_{in}} = \frac{r_z}{R + r_z}$$

For good line regulation, $r_z \ll R$, which ensures $S_V \ll 1$ (typically $S_V \approx 0.01 - 0.05$).

Final Verified Result

$$S_V = \frac{\Delta V_o}{\Delta V_{in}} = \frac{r_z}{R + r_z} \approx \frac{r_z}{R} \ll 1$$

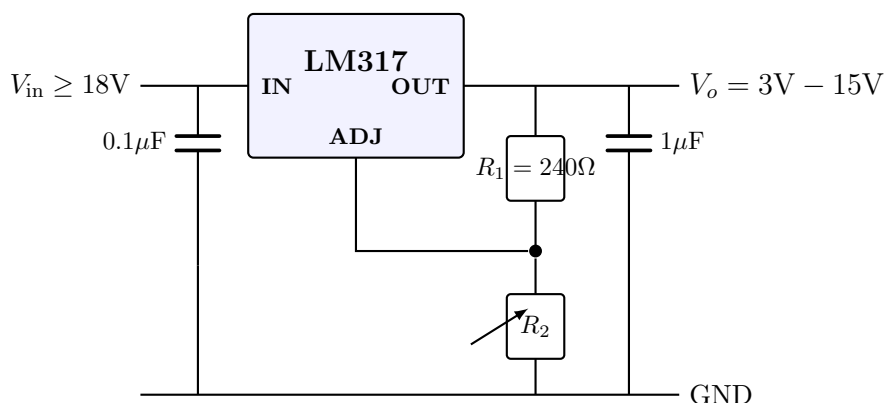
1.0.10 Question 10: LM317 Variable DC Voltage Regulator Design (3V - 15V) [4 Marks]**Problem Statement (2083 Baishakh, Q10)**

Design a variable DC voltage regulator using the LM317 to output (3-15) volts.

The LM317 maintains a precision internal reference voltage of $V_{ref} = 1.25$ V between its Output and Adjust terminals:

$$V_{out} = V_{ref} \left(1 + \frac{R_2}{R_1} \right) + I_{adj} R_2 \approx 1.25 \text{ V} \left(1 + \frac{R_2}{R_1} \right) \quad (\text{since } I_{adj} \approx 50 \mu\text{A} \text{ is negligible})$$

Circuit Schematic / Analysis Diagram: LM317 Adjustable Linear Voltage Regulator Circuit



Component Value Calculations:

1. Standard program resistor selection: $R_1 = 240 \Omega$ (satisfies $I_L \geq 5 \text{ mA}$ minimum load requirement).
2. For minimum output $V_{\text{out}(\min)} = 3.0 \text{ V}$:

$$3.0 = 1.25 \left(1 + \frac{R_{2(\min)}}{240} \right) \Rightarrow 1 + \frac{R_{2(\min)}}{240} = 2.4 \Rightarrow R_{2(\min)} = 1.4 \times 240 = \mathbf{336 \Omega}$$

3. For maximum output $V_{\text{out}(\max)} = 15.0 \text{ V}$:

$$15.0 = 1.25 \left(1 + \frac{R_{2(\max)}}{240} \right) \Rightarrow 1 + \frac{R_{2(\max)}}{240} = 12.0 \Rightarrow R_{2(\max)} = 11 \times 240 = \mathbf{2640 \Omega = 2.64 \text{ k}\Omega}$$

Practical Realization: Use a fixed resistor $R_{2A} = 330 \Omega$ in series with a **2.5 kΩ** linear potentiometer ($R_2 = 330 \Omega$ to 2.83 kΩ). Input DC voltage requirement: $V_{\text{in}} \geq V_{\text{out}(\max)} + 3 \text{ V} = 18 \text{ V DC}$.

Final Verified Result

$$\mathbf{R_1 = 240 \Omega, \quad R_2 = 336 \Omega \text{ to } 2.64 \text{ k}\Omega \quad (\text{Fixed } 330 \Omega + 2.5 \text{ k}\Omega \text{ Potentiometer}), \quad V_{\text{in}} \geq 18 \text{ V}}$$

Chapter 2

2082 Bhadra Examination Solutions

Examination Overview

Level: Bachelor in Engineering (BE)

Full Marks: 60

Programme: BEI, BCT

Pass Marks: 24

Year / Part: I / II

Time: 3 Hours

Subject: Electronic Devices & Circuits (ENEX 151)

Examination Type: Regular (New Course)

2.0.1 Question 1: β -Independent Voltage Divider Bias Design [1 + 5 = 6 Marks]

Problem Statement (2082 Bhadra, Q1)

What is the significance of β -independency in a voltage divider biasing circuit? Design a β -independent voltage divider bias circuit for the given parameters: $V_{CC} = 15\text{ V}$, $I_E = 1\text{ mA}$, and $\beta = 100$.

Part (a): Significance of β -Independency [1 Mark] Transistor forward current gain β (h_{FE}) exhibits wide manufacturing dispersions (typically ranging from 100 to 300 for the same transistor type) and strong temperature dependence ($+0.5\%/^{\circ}\text{C}$). A β -independent biasing circuit ensures that the quiescent operating point (I_{CQ}, V_{CEQ}) remains strictly invariant against device replacement and thermal drift, completely avoiding thermal runaway and waveform clipping.

Part (b): Step-by-Step Design [5 Marks] Design Specifications: $V_{CC} = 15\text{ V}$, $I_E \approx I_{CQ} = 1.0\text{ mA}$, $\beta = 100$, Silicon BJT ($V_{BE} = 0.7\text{ V}$).

1. **Emitter Voltage Allocation (V_E):** To ensure excellent temperature stability without compromising dynamic output voltage swing, allocate 10% of V_{CC} :

$$V_E = 0.10 \times V_{CC} = 0.10 \times 15\text{ V} = \mathbf{1.5\text{ V}}$$

2. Calculation of R_E :

$$R_E = \frac{V_E}{I_E} = \frac{1.5 \text{ V}}{1.0 \text{ mA}} = \mathbf{1.5 \text{ k}\Omega}$$

3. Collector Voltage Allocation (V_C): For maximum symmetrical undistorted output voltage swing:

$$V_{CEQ} = \frac{V_{CC} - V_E}{2} = \frac{15 \text{ V} - 1.5 \text{ V}}{2} = 6.75 \text{ V}$$

$$V_C = V_E + V_{CEQ} = 1.5 \text{ V} + 6.75 \text{ V} = 8.25 \text{ V}$$

$$R_C = \frac{V_{CC} - V_C}{I_C} = \frac{15 \text{ V} - 8.25 \text{ V}}{1.0 \text{ mA}} = 6.75 \text{ k}\Omega \quad (\text{Select standard value } R_C = \mathbf{6.8 \text{ k}\Omega})$$

4. Base Voltage (V_B):

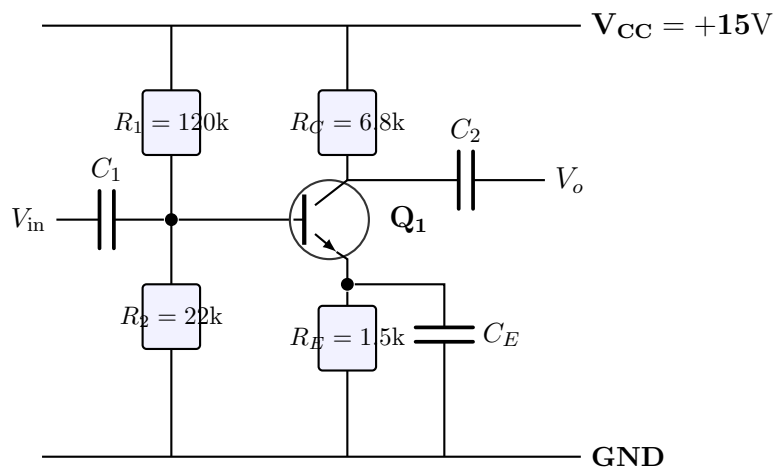
$$V_B = V_E + V_{BE} = 1.5 \text{ V} + 0.7 \text{ V} = \mathbf{2.2 \text{ V}}$$

5. Voltage Divider Resistors (R_1, R_2) via Bleeder Current Rule ($I_{\text{bleed}} \geq 10I_B$):

$$I_B = \frac{I_C}{\beta} = \frac{1.0 \text{ mA}}{100} = 0.01 \text{ mA} = 10 \mu\text{A} \implies I_{\text{bleed}} = 10 \times I_B = 0.10 \text{ mA}$$

$$R_2 = \frac{V_B}{I_{\text{bleed}}} = \frac{2.2 \text{ V}}{0.10 \text{ mA}} = \mathbf{22 \text{ k}\Omega} \quad (\text{Standard 5\% resistor})$$

$$R_1 = \frac{V_{CC} - V_B}{I_{\text{bleed}} + I_B} = \frac{15 \text{ V} - 2.2 \text{ V}}{0.10 \text{ mA} + 0.01 \text{ mA}} = \frac{12.8 \text{ V}}{0.11 \text{ mA}} = 116.36 \text{ k}\Omega \quad (\text{Select standard } R_1 = \mathbf{120 \text{ k}\Omega})$$

Circuit Schematic / Analysis Diagram: Designed Common Emitter Amplifier Circuit ($V_{CC} = 15\text{V}$)

Final Verified Result

$$\mathbf{R_1 = 120 \text{ k}\Omega, \quad R_2 = 22 \text{ k}\Omega, \quad R_C = 6.8 \text{ k}\Omega, \quad R_E = 1.5 \text{ k}\Omega}$$

2.0.2 Question 2: Role of Coupling Capacitors & CE Small Signal Parameters [2 + 5 = 7 Marks]

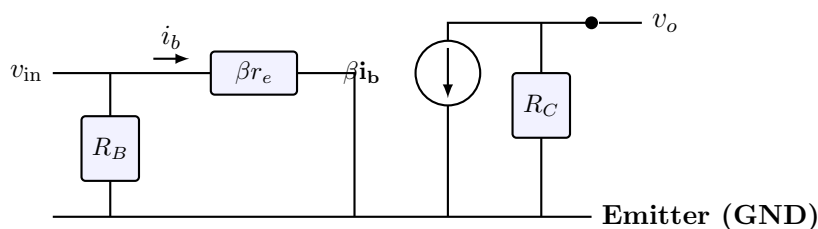
Problem Statement (2082 Bhadra, Q2)

Explain the role of coupling capacitors in amplifier circuits. Draw the small signal equivalent circuit of a common emitter amplifier circuit and find its input impedance, output impedance and voltage gain.

Part (a): Role of Coupling Capacitors (C_{in}, C_{out}) [2 Marks] Coupling capacitors block DC operating voltages from entering the external AC signal source or load resistor, preventing quiescent operating point shifts while presenting a negligible impedance path ($X_C = \frac{1}{2\pi fC} \approx 0$) for signal frequencies.

Part (b): Common Emitter Parameter Derivations [5 Marks]

Circuit Schematic / Analysis Diagram: Small-Signal Model of Common Emitter BJT Amplifier



1. Input Impedance (Z_{in}):

$$Z_{in} = R_B \parallel \beta r_e \approx \beta r_e \quad (\text{moderate, } 1 \text{ k}\Omega - 5 \text{ k}\Omega)$$

2. Output Impedance (Z_{out}):

$$Z_{out} = R_C \parallel r_o \approx R_C$$

3. Voltage Gain (A_v):

$$v_o = -(\beta i_b)(R_C \parallel R_L), \quad v_{in} = i_b(\beta r_e)$$

$$A_v = \frac{v_o}{v_{in}} = -\frac{\beta i_b(R_C \parallel R_L)}{i_b \beta r_e} = -\frac{R_C \parallel R_L}{r_e} \quad (180^\circ \text{ phase inversion})$$

4. Current Gain (A_i):

$$A_i = \frac{i_o}{i_{in}} \approx -\beta \left(\frac{R_B}{R_B + \beta r_e} \right) \left(\frac{R_C}{R_C + R_L} \right)$$

Final Verified Result

$$Z_{in} \approx R_B \parallel \beta r_e, \quad Z_{out} \approx R_C, \quad A_v = -\frac{R_C \parallel R_L}{r_e}, \quad A_i \approx -\beta$$

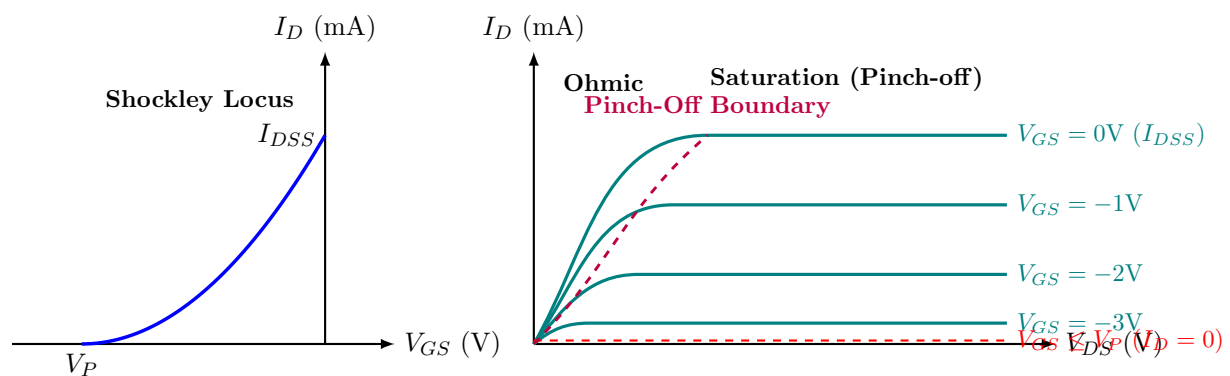
2.0.3 Question 3: N-Channel JFET Construction and Characteristics [6 Marks]

Problem Statement (2082 Bhadra, Q3)

Describe the construction and working principle of n-channel JFET with the help of characteristic curves and mathematical expressions.

1. Physical Construction: An N-Channel Junction Field-Effect Transistor consists of an n-type semiconductor silicon bar with two ohmic contacts at the ends forming the **Drain** and **Source**. Two heavily doped p^+ regions are diffused on opposite sides of the bar and internally connected to form the **Gate**.

Circuit Schematic / Analysis Diagram: Transfer Characteristics and Drain Output Curves of N-Channel JFET



2. Conduction Equations:

- Ohmic Region ($V_{DS} < V_{GS} - V_P$):

$$I_D = I_{DSS} \left[2 \left(1 - \frac{V_{GS}}{V_P} \right) \frac{V_{DS}}{|V_P|} - \left(\frac{V_{DS}}{V_P} \right)^2 \right]$$

- Saturation Region ($V_{DS} \geq V_{GS} - V_P$):

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2, \quad g_m = \frac{2I_{DSS}}{|V_P|} \left(1 - \frac{V_{GS}}{V_P} \right)$$

Final Verified Result

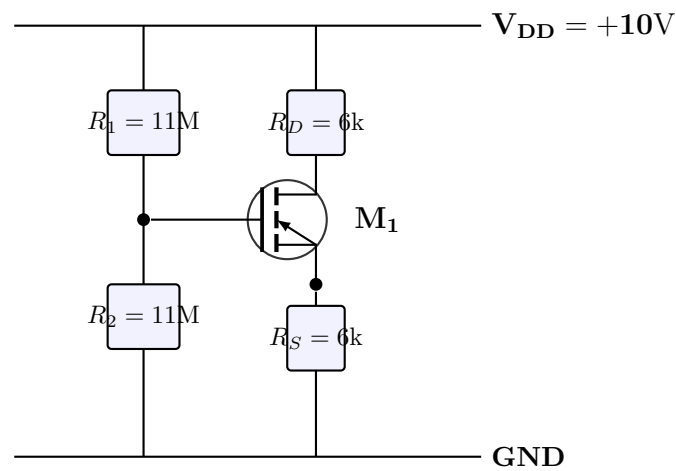
$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2, \quad V_{DS(sat)} = V_{GS} - V_P$$

2.0.4 Question 4: N-Channel Enhancement-Type MOSFET Q-Point Calculation [7 Marks]

Problem Statement (2082 Bhadra, Q4)

Find I_D and V_{DS} for the circuit shown in the figure below. Given parameters are: $V_t = 1$ V and $K = 0.48$ mA/V². Circuit values: $V_{DD} = +10$ V, $R_1 = 11$ M Ω , $R_2 = 11$ M Ω , $R_D = 6$ k Ω , $R_S = 6$ k Ω .

Circuit Schematic / Analysis Diagram: N-Channel E-MOSFET Voltage Divider Bias Circuit



1. Gate DC Voltage (V_G): Since the insulated gate draws zero current ($I_G = 0$):

$$V_G = V_{DD} \left(\frac{R_2}{R_1 + R_2} \right) = 10 \text{ V} \times \left(\frac{11 \text{ M}\Omega}{11 \text{ M}\Omega + 11 \text{ M}\Omega} \right) = 5.0 \text{ V}$$

2. Equation for V_{GS} : Applying KVL in the gate-source loop:

$$V_{GS} = V_G - I_D R_S = 5.0 - 6.0 I_D \quad (\text{with } I_D \text{ in mA and } R_S = 6 \text{ k}\Omega)$$

3. Saturation Region Characteristic Equation:

$$I_D = K(V_{GS} - V_t)^2 = 0.48 ((5.0 - 6.0 I_D) - 1.0)^2 = 0.48(4.0 - 6.0 I_D)^2$$

$$I_D = 0.48(16.0 - 48.0 I_D + 36.0 I_D^2) = 7.68 - 23.04 I_D + 17.28 I_D^2$$

$$17.28 I_D^2 - 24.04 I_D + 7.68 = 0$$

Solving via the quadratic formula:

$$\Delta = (-24.04)^2 - 4(17.28)(7.68) = 577.9216 - 530.8416 = 47.0800$$

$$\sqrt{\Delta} = \sqrt{47.0800} = 6.8615$$

$$I_D = \frac{24.04 \pm 6.8615}{2(17.28)} = \frac{24.04 \pm 6.8615}{34.56}$$

Two mathematical roots:

- $I_{D1} = \frac{30.9015}{34.56} = 0.8941 \text{ mA} \implies V_{GS1} = 5.0 - 6(0.8941) = -0.365 \text{ V} < V_t = 1.0 \text{ V}$ (**Invalid, below threshold**).
- $I_{D2} = \frac{17.1785}{34.56} = \mathbf{0.4971 \text{ mA}} \implies V_{GS2} = 5.0 - 6(0.4971) = \mathbf{2.017 \text{ V}} > V_t = 1.0 \text{ V}$ (**Valid, transistor ON**).

4. Calculation of Drain-to-Source Voltage (V_{DS}):

$$V_{DS} = V_{DD} - I_D(R_D + R_S) = 10 \text{ V} - 0.4971 \text{ mA} \times (6 \text{ k}\Omega + 6 \text{ k}\Omega) = 10 - 0.4971(12.0) = \mathbf{4.035 \text{ V}}$$

Pinch-Off (Saturation) Verification:

$$V_{DS} = 4.035 \text{ V} \geq V_{GS} - V_t = 2.017 \text{ V} - 1.0 \text{ V} = 1.017 \text{ V} \implies \text{Confirmed in Saturation Region! } \checkmark$$

Final Verified Result

$$I_D = \mathbf{0.497 \text{ mA}}, \quad V_{GS} = \mathbf{2.017 \text{ V}}, \quad V_{DS} = \mathbf{4.035 \text{ V}}$$

2.0.5 Question 5: Barkhausen Criteria & RC Phase Shift Oscillator [6 Marks]

Problem Statement (2082 Bhadra, Q5)

State Barkhausen criteria for sinusoidal oscillation. Explain the working principle of RC phase shift oscillator with necessary expression and circuit diagram.

Part (a): Barkhausen Criteria for Sustained Sinusoidal Oscillation [2 Marks] For a closed-loop feedback amplifier to sustain stable sinusoidal oscillations without an external input:

1. **Loop Gain Magnitude Criterion:** The magnitude of the open-loop gain must be at least unity:

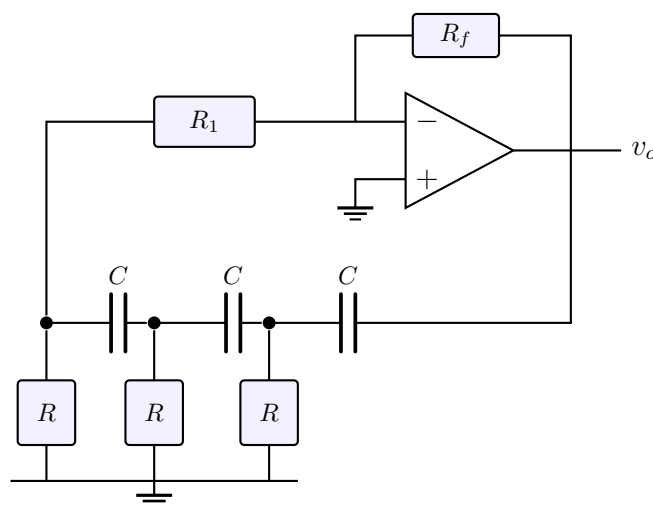
$$|\mathbf{A}\beta| \geq 1$$

2. **Loop Phase Shift Criterion:** The total phase shift around the closed loop must be an integer multiple of 360° (0°):

$$\angle(\mathbf{A}\beta) = 0^\circ \quad \text{or} \quad 360^\circ$$

Part (b): Working Principle of Op-Amp RC Phase Shift Oscillator [4 Marks]

Circuit Schematic / Analysis Diagram: Op-Amp RC Phase Shift Oscillator Circuit



Mathematical Derivation of f_0 and Gain Requirement: The transfer function of the 3-stage high-pass RC ladder feedback network is:

$$\beta(s) = \frac{v_f(s)}{v_o(s)} = \frac{s^3 R^3 C^3}{s^3 R^3 C^3 + 6s^2 R^2 C^2 + 5sRC + 1}$$

Substituting $s = j\omega$:

$$\beta(j\omega) = \frac{-j\omega^3 R^3 C^3}{(1 - 6\omega^2 R^2 C^2) + j(5\omega RC - \omega^3 R^3 C^3)}$$

- Oscillation Frequency (f_0):** For the feedback network to produce a 180° phase shift, the real part in the denominator must vanish:

$$1 - 6\omega_0^2 R^2 C^2 = 0 \implies \omega_0 = \frac{1}{RC\sqrt{6}} \implies f_0 = \frac{1}{2\pi RC\sqrt{6}}$$

- Feedback Attenuation at f_0 :**

$$\beta(j\omega_0) = \frac{-j(1/6\sqrt{6})}{j(5/\sqrt{6} - 1/6\sqrt{6})} = \frac{-1/6}{29/6} = -\frac{1}{29} \implies |\beta| = \frac{1}{29}$$

- Minimum Amplifier Gain ($|A_v\beta| \geq 1$):**

$$|A_v| = \frac{R_f}{R_1} \geq \frac{1}{|\beta|} = 29 \implies R_f \geq 29R_1$$

Final Verified Result

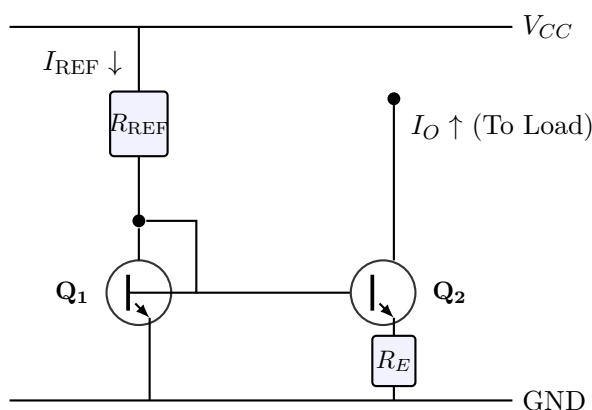
$$|A\beta| \geq 1, \quad \angle(A\beta) = 360^\circ, \quad f_0 = \frac{1}{2\pi RC\sqrt{6}}, \quad |A_v| \geq 29 \implies R_f \geq 29R_1$$

2.0.6 Question 6: Widlar Current Source Operation & Advantages [5 + 2 = 7 Marks]

Problem Statement (2082 Bhadra, Q6)

With a neat circuit diagram and necessary mathematical expressions, explain the operation of Widlar current source. What is the advantage of Widlar current source over simple current source?

Circuit Schematic / Analysis Diagram: BJT Widlar Current Source Circuit



Mathematical Analysis:

1. Applying KVL around the base-emitter loop of Q_1 and Q_2 :

$$V_{BE1} = V_{BE2} + I_{E2}R_E \approx V_{BE2} + I_O R_E \implies I_O R_E = V_{BE1} - V_{BE2}$$

2. Expressing base-emitter voltages using the Ebers-Moll equation $I_C = I_S e^{V_{BE}/V_T}$:

$$V_{BE1} = V_T \ln \left(\frac{I_{REF}}{I_{S1}} \right), \quad V_{BE2} = V_T \ln \left(\frac{I_O}{I_{S2}} \right)$$

Assuming matched transistors ($I_{S1} = I_{S2}$):

$$I_O R_E = V_T \ln \left(\frac{I_{REF}}{I_O} \right) \implies R_E = \frac{V_T}{I_O} \ln \left(\frac{I_{REF}}{I_O} \right)$$

Part (b): Advantages over Simple Current Source [2 Marks]

- **Micro-Ampere Current Generation ($I_O \ll I_{REF}$):** Enables setting small bias currents (μA range) using small integrated resistors ($\sim k\Omega$) rather than huge, silicon-area-prohibitive mega-ohm resistors.
- **Ultra-High Output Resistance:** Negative current feedback across R_E increases output impedance: $R_{out} \approx r_{o2}(1 + g_{m2}R_E)$.

Final Verified Result

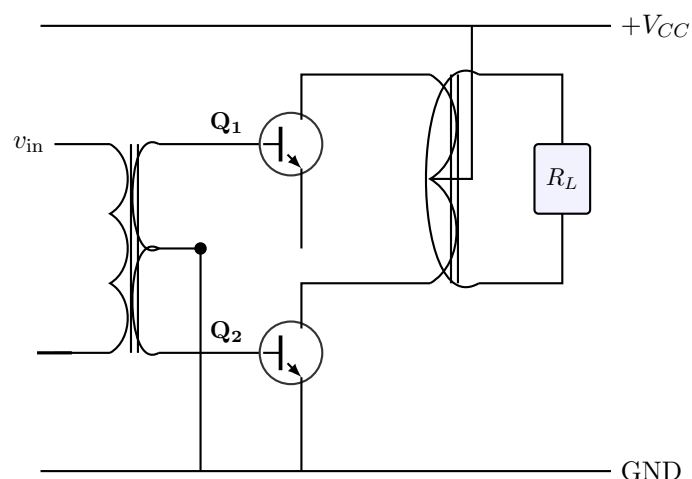
$$I_O R_E = V_T \ln \left(\frac{I_{REF}}{I_O} \right), \quad R_{out} \approx r_{o2}(1 + g_{m2} R_E)$$

2.0.7 Question 7: Transformer-Coupled Class B Amplifier & Minimum Efficiency [6 Marks]

Problem Statement (2082 Bhadra, Q7)

Draw the circuit diagram of transformer coupled Class B amplifier. Determine the condition for minimum efficiency of Class B amplifier.

Circuit Schematic / Analysis Diagram: Transformer-Coupled Class B Push-Pull Amplifier Circuit



Efficiency Equation and Minimum Efficiency Condition:

1. General Efficiency Expression:

$$P_{o(ac)} = \frac{V_p^2}{2R'_L}, \quad P_{i(dc)} = \frac{2}{\pi} \frac{V_{CC} V_p}{R'_L} \Rightarrow \eta = \frac{\pi}{4} \left(\frac{V_p}{V_{CC}} \right) \times 100\%$$

2. Condition for Minimum Efficiency (η_{min}):

Since efficiency is directly proportional to the output voltage amplitude V_p :

$$\text{As } V_p \rightarrow 0 \Rightarrow \eta_{min} = 0\%$$

Minimum conversion efficiency ($\eta = 0\%$) occurs at ****zero input signal amplitude**** (quiescent no-signal condition).

3. Condition for Maximum Efficiency (η_{max}):

Occurs when the output signal reaches maximum

undistorted swing ($V_p = V_{CC}$):

$$\eta_{\max} = \frac{\pi}{4} \times 100\% \approx 78.54\%$$

Final Verified Result

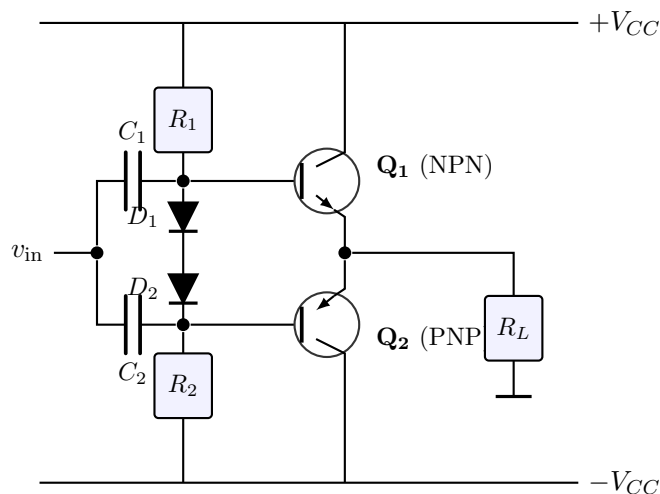
$$\eta = \frac{\pi}{4} \left(\frac{V_p}{V_{CC}} \right) \times 100\%, \quad \eta_{\min} = 0\% \quad (\text{at } V_p = 0), \quad \eta_{\max} = 78.54\% \quad (\text{at } V_p = V_{CC})$$

2.0.8 Question 8: Crossover Distortion & Class AB Elimination [6 Marks]

Problem Statement (2082 Bhadra, Q8)

Define crossover distortion in a class B amplifier and explain how class AB operation can be used to eliminate it.

Circuit Schematic / Analysis Diagram: Class AB Complementary Push-Pull Amplifier with Diode Biasing



1. Definition of Crossover Distortion: In a standard Class B push-pull stage, transistors remain unconducting whenever the input signal magnitude is below the base-emitter cut-in voltage ($|v_{in}| < 0.7 \text{ V}$). This causes a flat zero-crossing interval where both transistors are completely OFF, generating severe non-linear harmonic distortion termed ****crossover distortion****.

2. Elimination Mechanism via Class AB Biasing:

1. Diodes D_1 and D_2 establish a constant forward bias voltage $V_{BB} = V_{D1} + V_{D2} \approx 1.4 \text{ V} \approx 2V_{BE}$ across the bases of Q_1 and Q_2 .

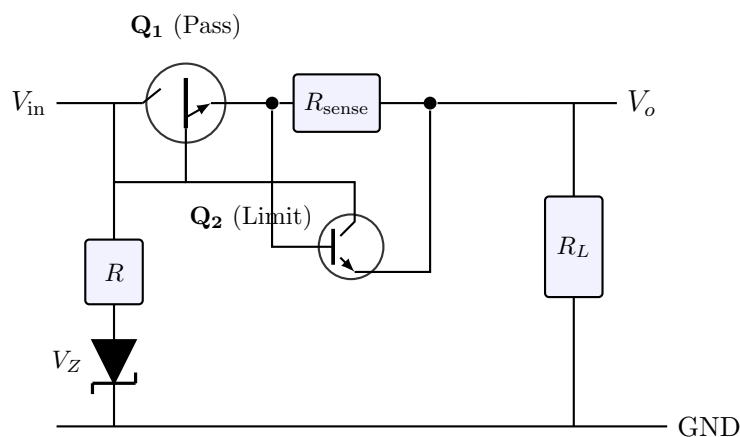
2. This trickles a small quiescent collector current ($I_{CQ} \approx 5 \text{ mA} - 20 \text{ mA}$) through both transistors at $v_{in} = 0 \text{ V}$.
3. Consequently, transistors transition smoothly without any zero-crossing dead-band interval, completely eliminating crossover distortion.

Final Verified Result

$$V_{BB} = V_{D1} + V_{D2} \approx 2V_{BE} \Rightarrow \text{Completely Eliminates Zero-Crossing Dead-Band Distortion}$$

2.0.9 Question 9: Series Transistor Regulator with Current Limiting [5 Marks]**Problem Statement (2082 Bhadra, Q9)**

Explain the working principle of a series transistor voltage regulator, accompanied by short-circuit/overload protection.

Circuit Schematic / Analysis Diagram: Series Transistor Voltage Regulator with Short-Circuit Current Limiting Protection**Working Principle and Current Limit Derivation:**

1. The load current I_L flows directly through R_{sense} , developing a voltage drop $V_{sense} = I_L R_{sense} = V_{BE2}$.
2. **Under Normal Conditions** ($I_L < I_{limit}$): $V_{BE2} < 0.6 \text{ V} \Rightarrow$ Protection transistor Q_2 remains completely OFF and does not affect the output voltage: $V_o \approx V_Z - V_{BE1}$.
3. **Under Overload / Short-Circuit Fault** ($I_L \geq I_{limit}$): When V_{sense} reaches the cut-in threshold ($V_{BE2} \approx 0.7 \text{ V}$), transistor Q_2 turns ON and begins shunting excess base drive current away from pass transistor Q_1 .
4. Any attempt to draw more current causes Q_2 to conduct harder, throttling Q_1 and clamping the

load current strictly at:

$$I_{\text{limit}} = \frac{V_{\text{BE2(on)}}}{R_{\text{sense}}} \approx \frac{0.7 \text{ V}}{R_{\text{sense}}}$$

Final Verified Result

$$I_{\text{limit}} = \frac{V_{\text{BE2}}}{R_{\text{sense}}} \approx \frac{0.7 \text{ V}}{R_{\text{sense}}}$$

2.0.10 Question 10: LM317 Adjustable Regulator Design (10V - 25V) [4 Marks]

Problem Statement (2082 Bhadra, Q10)

Design a 10-25 V variable DC voltage regulator using LM317 IC.

$$V_{\text{out}} = 1.25 \text{ V} \left(1 + \frac{R_2}{R_1} \right)$$

1. Select standard program resistor $R_1 = 240 \Omega$.

2. For $V_{\text{out(min)}} = 10.0 \text{ V}$:

$$10.0 = 1.25 \left(1 + \frac{R_{2(\text{min})}}{240} \right) \Rightarrow 1 + \frac{R_{2(\text{min})}}{240} = 8.0 \Rightarrow R_{2(\text{min})} = 7 \times 240 = 1680 \Omega = 1.68 \text{ k}\Omega$$

3. For $V_{\text{out(max)}} = 25.0 \text{ V}$:

$$25.0 = 1.25 \left(1 + \frac{R_{2(\text{max})}}{240} \right) \Rightarrow 1 + \frac{R_{2(\text{max})}}{240} = 20.0 \Rightarrow R_{2(\text{max})} = 19 \times 240 = 4560 \Omega = 4.56 \text{ k}\Omega$$

Practical Realization: Use a fixed precision resistor $R_{2A} = 1.6 \text{ k}\Omega$ in series with a **3.0 k** Ω linear potentiometer ($R_2 = 1.68 \text{ k}\Omega$ to $4.56 \text{ k}\Omega$). Input DC voltage: $V_{\text{in}} \geq 28 \text{ V DC}$.

Final Verified Result

$$R_1 = 240 \Omega, \quad R_2 = 1.68 \text{ k}\Omega \text{ to } 4.56 \text{ k}\Omega \quad (\text{Fixed } 1.6 \text{ k}\Omega + 3.0 \text{ k}\Omega \text{ Potentiometer})$$

Chapter 3

2082 Baishakh Examination Solutions

Examination Overview

Level: Bachelor in Engineering (BE)

Full Marks: 60

Programme: BEI, BCT

Pass Marks: 24

Year / Part: I / II

Time: 3 Hours

Subject: Electronic Devices & Circuits (EX 151)

Examination Type: Back (New Course)

3.0.1 Question 1: β -Independent Voltage Divider Bias Design [4 Marks]

Problem Statement (2082 Baishakh, Q1)

Design a β -independent voltage divider bias circuit using the appropriate guidelines for the given parameters: $V_{CC} = 12\text{V}$, $I_C = 1\text{mA}$, and $\beta = 100$.

1. **Emitter Voltage Allocation (V_E):** Set $V_E = 0.10 \times V_{CC} = 0.10 \times 12\text{V} = 1.2\text{ V}$.

$$R_E = \frac{V_E}{I_E} \approx \frac{1.2\text{V}}{1.0\text{mA}} = 1.2\text{ k}\Omega$$

2. **Collector Resistor Allocation (R_C):** Set $V_{CEQ} = \frac{V_{CC} - V_E}{2} = \frac{12 - 1.2}{2} = 5.4\text{V}$:

$$R_C = \frac{V_{CC} - (V_E + V_{CEQ})}{I_C} = \frac{12 - 6.6}{1.0\text{mA}} = 5.4\text{ k}\Omega \quad (\text{Select standard } R_C = 5.6\text{ k}\Omega)$$

3. **Base Voltage Calculation (V_B):**

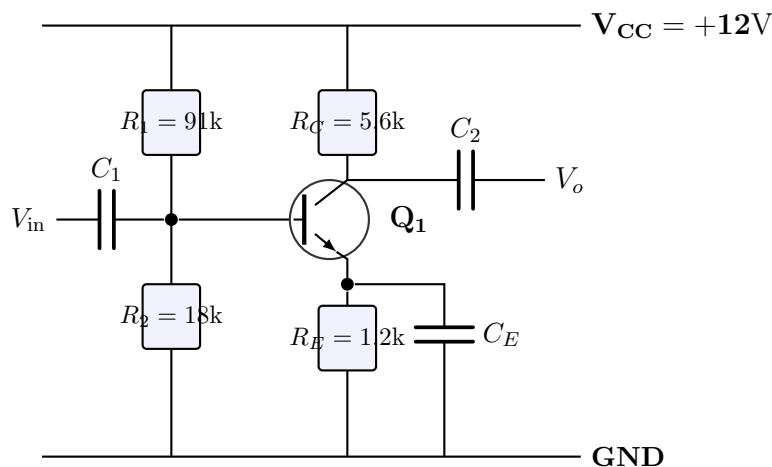
$$V_B = V_E + V_{BE} = 1.2\text{V} + 0.7\text{V} = 1.9\text{ V}$$

4. Stiff Biasing Voltage Divider Calculation ($I_{\text{bleed}} \geq 10I_B$):

$$I_B = \frac{1\text{mA}}{100} = 10 \mu\text{A} \implies I_{\text{bleed}} = 100 \mu\text{A} = 0.10\text{mA}$$

$$R_2 = \frac{V_B}{I_{\text{bleed}}} = \frac{1.9\text{V}}{0.10\text{mA}} = 19 \text{ k}\Omega \quad (\text{Standard } R_2 = 18 \text{ k}\Omega \text{ or } 20 \text{ k}\Omega)$$

$$R_1 = \frac{V_{CC} - V_B}{I_{\text{bleed}} + I_B} = \frac{12\text{V} - 1.9\text{V}}{0.11\text{mA}} = \frac{10.1\text{V}}{0.11\text{mA}} = 91.8 \text{ k}\Omega \quad (\text{Standard } R_1 = 91 \text{ k}\Omega)$$

Circuit Schematic / Analysis Diagram: Designed Common Emitter Amplifier Schematic ($V_{CC} = 12\text{V}$)

Final Verified Result

$$R_1 = 91 \text{ k}\Omega, \quad R_2 = 18 \text{ k}\Omega, \quad R_C = 5.6 \text{ k}\Omega, \quad R_E = 1.2 \text{ k}\Omega$$

3.0.2 Question 2: CE BJT Amplifier Small Signal Parameter Calculations [5 Marks]

Problem Statement (2082 Baishakh, Q2)

Determine input resistance, voltage gain and output resistance of given common emitter BJT Amplifier. $\beta = 120$ and $r_o = 40 \text{ k}\Omega$. Circuit components: $V_{CC} = 20\text{V}$, $R_1 = 470 \text{ k}\Omega$, $R_2 = 56 \text{ k}\Omega$, $R_C = 2.2 \text{ k}\Omega$, $R_E = 0.56 \text{ k}\Omega$ (with bypass capacitor C_E).

1. DC Bias Operating Point Analysis:

- Thevenin Base Voltage:

$$V_{\text{th}} = V_{CC} \left(\frac{R_2}{R_1 + R_2} \right) = 20\text{V} \times \left(\frac{56\text{k}\Omega}{470\text{k}\Omega + 56\text{k}\Omega} \right) = 20 \times \frac{56}{526} = 2.129 \text{ V}$$

- Thevenin Resistance:

$$R_{th} = R_1 \parallel R_2 = \frac{470 \times 56}{526} = \mathbf{50.038 \text{ k}\Omega}$$

- Quiescent Base and Emitter Currents:

$$I_B = \frac{V_{th} - V_{BE}}{R_{th} + (1 + \beta)R_E} = \frac{2.129\text{V} - 0.7\text{V}}{50.038\text{k}\Omega + (121)(0.56\text{k}\Omega)} = \frac{1.429\text{V}}{50.038\text{k}\Omega + 67.76\text{k}\Omega} = \frac{1.429\text{V}}{117.8\text{k}\Omega} = \mathbf{12.13 \mu\text{A}}$$

$$I_E = (1 + \beta)I_B = 121 \times 12.13 \mu\text{A} = \mathbf{1.468 \text{ mA}}$$

- Dynamic Emitter Resistance:

$$r_e = \frac{V_T}{I_E} = \frac{26 \text{ mV}}{1.468 \text{ mA}} = \mathbf{17.71 \Omega}$$

2. Small-Signal AC Parameters:

1. Input Resistance (R_{in}):

$$Z_b = \beta r_e = 120 \times 17.71 \Omega = \mathbf{2.125 \text{ k}\Omega}$$

$$R_{in} = R_1 \parallel R_2 \parallel Z_b = 50.038\text{k}\Omega \parallel 2.125\text{k}\Omega = \frac{50.038 \times 2.125}{52.163} = \mathbf{2.038 \text{ k}\Omega}$$

2. Output Resistance (R_{out}):

$$R_{out} = R_C \parallel r_o = 2.2\text{k}\Omega \parallel 40\text{k}\Omega = \frac{2.2 \times 40}{42.2} = \mathbf{2.085 \text{ k}\Omega}$$

3. Voltage Gain (A_v):

$$A_v = -\frac{R_C \parallel r_o}{r_e} = -\frac{2085.3 \Omega}{17.71 \Omega} = \mathbf{-117.75} \quad (180^\circ \text{ phase inversion})$$

Final Verified Result

$$\mathbf{R_{in} = 2.038 \text{ k}\Omega, \quad R_{out} = 2.085 \text{ k}\Omega, \quad A_v = -117.75}$$

3.0.3 Question 3: BJT Operation as a Switch & Dynamic Switching Times [4 Marks]

Problem Statement (2082 Baishakh, Q3)

Explain the operation principle of a BJT as a switch with the necessary diagrams.

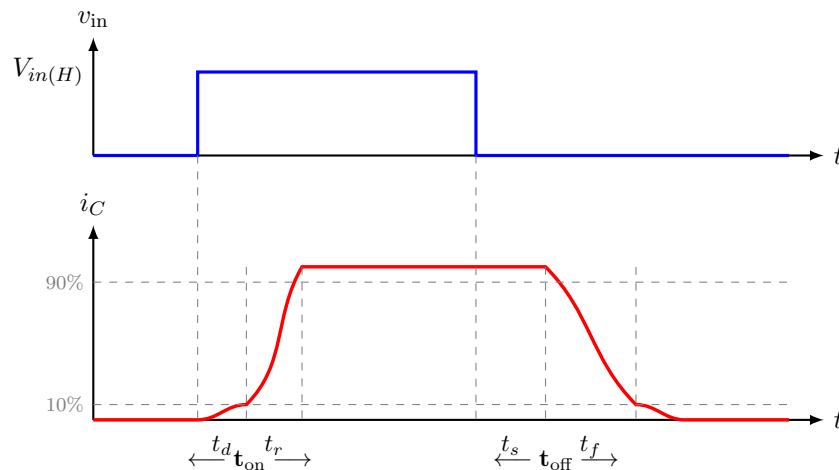
1. Operating States of BJT as a Switch:

- **OFF State (Cutoff Region):** $V_{in} \leq 0\text{V} \implies I_B = 0 \implies I_C \approx 0$. Transistor behaves as an open contact; $V_{CE} = V_{CC}$.

- **ON State (Saturation Region):** $V_{in} \gg V_{BE} \implies I_B \geq I_{B(sat)} = \frac{I_{C(sat)}}{\beta_{min}}$. Transistor behaves as a closed contact; $V_{CE} = V_{CE(sat)} \approx 0.1V - 0.2V$.

2. Transient Switching Times:

Circuit Schematic / Analysis Diagram: BJT Dynamic Switching Waveforms and Interval Definitions



- **Turn-On Time** ($t_{on} = t_d + t_r$): Delay time t_d (junction capacitance charging) + Rise time t_r (10% to 90% of I_C).
- **Turn-Off Time** ($t_{off} = t_s + t_f$): Storage time t_s (removal of excess stored base charge) + Fall time t_f (90% down to 10%).

Final Verified Result

BJT acts as an inverter switch between Cutoff ($V_{CE} = V_{CC}$) and Saturation ($V_{CE} \approx 0.2V$), with total switching speed governed by $t_{on} = t_d + t_r$ and $t_{off} = t_s + t_f$.

3.0.4 Question 4: N-Channel Depletion-Type MOSFET [7 Marks]

Problem Statement (2082 Baishakh, Q4)

Describe the construction and working principle of n-channel depletion type MOSFET with the help of characteristics curve and mathematical expressions.

1. Physical Construction: An N-Channel Depletion-Type MOSFET (D-MOSFET) is built on a p-type silicon substrate:

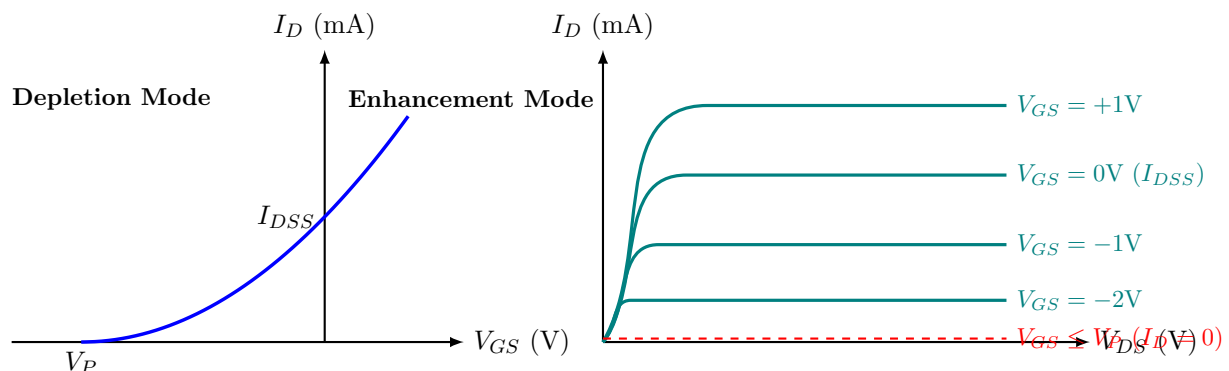
- Two heavily doped n^+ regions form the **Source** and **Drain**.

- A physical, chemically doped **n-type channel** is pre-diffused between source and drain during fabrication.
- The **Gate** terminal is completely insulated from the channel by a dielectric layer of Silicon Dioxide (SiO_2).

2. Working Principle:

1. **Depletion Mode** ($V_{GS} < 0$): Applying a negative gate voltage repels conduction electrons from the n-channel into the p-substrate, creating a depletion region and reducing drain current I_D . At $V_{GS} = V_P$ (Pinch-off voltage), the channel is fully depleted and $I_D = 0$.
2. **Enhancement Mode** ($V_{GS} > 0$): Applying a positive gate voltage attracts additional free electrons from the substrate into the channel, enhancing channel conductivity and increasing I_D above I_{DSS} .

Circuit Schematic / Analysis Diagram: Transfer and Drain Characteristic Curves of N-Channel D-MOSFET



3. **Mathematical Model (Shockley's Equation):** In the saturation (active) region ($V_{DS} \geq V_{GS} - V_P$):

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2 \quad \text{valid for both } V_{GS} \leq 0 \text{ and } V_{GS} > 0$$

Final Verified Result

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2, \quad g_m = \frac{2I_{DSS}}{|V_P|} \left(1 - \frac{V_{GS}}{V_P} \right)$$

3.0.5 Question 5: JFET Voltage Divider Bias Q-Point & Transconductance [6 Marks]

Problem Statement (2082 Baishakh, Q5)

For the given JFET voltage divider configuration, determine I_{DQ} , V_{GSQ} , V_{DS} and transconductance g_m . Given parameters: $V_{DD} = 20\text{V}$, $R_1 = 910\text{ k}\Omega$, $R_2 = 110\text{ k}\Omega$, $R_D = 2.2\text{ k}\Omega$, $R_S = 1.1\text{ k}\Omega$, $I_{DSS} = 10\text{mA}$, $V_P = -3.5\text{V}$.

1. Gate Voltage Calculation (V_G):

$$V_G = V_{DD} \left(\frac{R_2}{R_1 + R_2} \right) = 20\text{V} \times \left(\frac{110\text{k}\Omega}{910\text{k}\Omega + 110\text{k}\Omega} \right) = 20 \times \frac{110}{1020} = \mathbf{2.1569\text{ V}}$$

2. Equation for V_{GS} :

$$V_{GS} = V_G - I_D R_S = 2.1569 - 1.1 I_D \quad (\text{with } I_D \text{ in mA})$$

3. Substitution into Shockley's Law:

$$I_D = 10 \left(1 - \frac{2.1569 - 1.1 I_D}{-3.5} \right)^2 = 10 \left(\frac{3.5 + 2.1569 - 1.1 I_D}{3.5} \right)^2 = \frac{10}{12.25} (5.6569 - 1.1 I_D)^2$$

$$1.225 I_D = 32.0 - 12.445 I_D + 1.21 I_D^2 \implies 1.21 I_D^2 - 13.67 I_D + 32.0 = 0$$

Solving via quadratic formula:

$$I_D = \frac{13.67 \pm \sqrt{(13.67)^2 - 4(1.21)(32.0)}}{2(1.21)} = \frac{13.67 \pm \sqrt{186.87 - 154.88}}{2.42} = \frac{13.67 \pm 5.656}{2.42}$$

Two roots:

- $I_{D1} = 7.986\text{ mA} \implies V_{GS1} = 2.1569 - 1.1(7.986) = -6.628\text{V} < V_P$ (**Invalid**).
- $I_{D2} = \mathbf{3.312\text{ mA}} \implies V_{GS2} = 2.1569 - 1.1(3.312) = \mathbf{-1.486\text{ V}}$ (**Valid**, since $-3.5\text{V} \leq V_{GS} \leq 0$).

4. Calculation of V_{DS} and Transconductance g_m :

$$V_{DS} = V_{DD} - I_D(R_D + R_S) = 20\text{V} - 3.312\text{mA} \times (2.2\text{k}\Omega + 1.1\text{k}\Omega) = 20 - 3.312(3.3) = \mathbf{9.07\text{ V}}$$

$$g_m = \frac{2I_{DSS}}{|V_P|} \left(1 - \frac{V_{GSQ}}{V_P} \right) = \frac{2 \times 10\text{mA}}{3.5\text{V}} \left(1 - \frac{-1.486\text{V}}{-3.5\text{V}} \right) = 5.714 \times (1 - 0.4246) = \mathbf{3.288\text{ mS}}$$

Final Verified Result

$$I_{DQ} = \mathbf{3.312\text{ mA}}, \quad V_{GSQ} = \mathbf{-1.486\text{ V}}, \quad V_{DS} = \mathbf{9.07\text{ V}}, \quad g_m = \mathbf{3.288\text{ mS}}$$

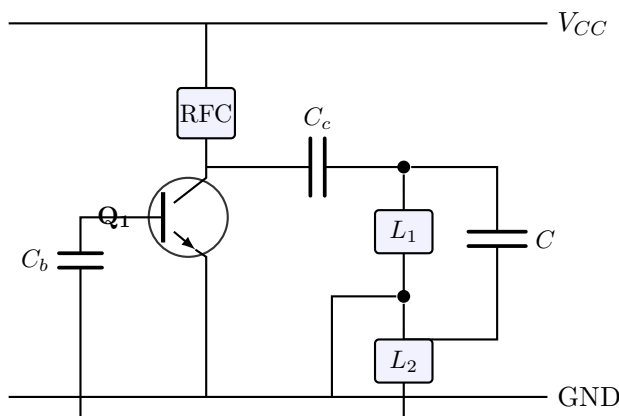
3.0.6 Question 6: LC Hartley Oscillator Frequency Derivation [4 Marks]

Problem Statement (2082 Baishakh, Q6)

Draw circuit diagram of LC Hartley oscillator. Derive its frequency of oscillation.

A Hartley oscillator employs an inductive voltage divider with two tapped inductors (L_1, L_2) having mutual inductance M , tuned by capacitor C .

Circuit Schematic / Analysis Diagram: BJT LC Hartley Oscillator Circuit



Derivation of Oscillation Frequency: The feedback factor is given by $\beta = \frac{X_{L2}}{X_{L1}} = \frac{L_2 + M}{L_1 + M}$. The equivalent tank inductance of the series tapped inductor is:

$$L_{eq} = L_1 + L_2 + 2M$$

The resonant frequency where inductive and capacitive reactances cancel ($X_{L_{eq}} = X_C$) is:

$$\omega_0 L_{eq} = \frac{1}{\omega_0 C} \Rightarrow \omega_0^2 = \frac{1}{L_{eq} C} \Rightarrow f_0 = \frac{1}{2\pi \sqrt{(L_1 + L_2 + 2M)C}}$$

Final Verified Result

$$f_0 = \frac{1}{2\pi \sqrt{(L_1 + L_2 + 2M)C}}$$

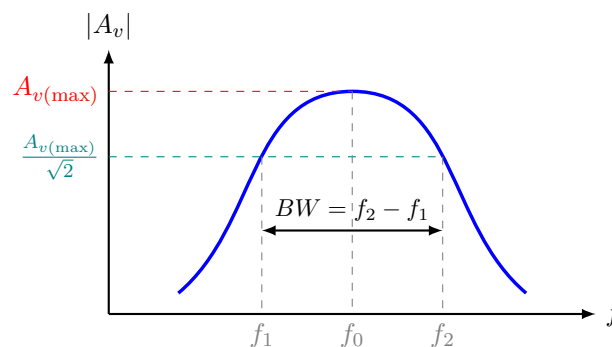
3.0.7 Question 7: Bandwidth of Class A Tuned Amplifier [4 Marks]

Problem Statement (2082 Baishakh, Q7)

Derive the expression of bandwidth of class A tuned amplifier.

A tuned amplifier uses a parallel LC tank circuit as its collector load. At the resonant frequency $f_0 = \frac{1}{2\pi\sqrt{LC}}$, the impedance is purely resistive: $R_p = Q^2 R_s = \frac{L}{CR_s}$.

Circuit Schematic / Analysis Diagram: Frequency Response Curve of Tuned Amplifier



- The 3dB half-power frequencies (f_1 and f_2) occur where the tank impedance falls to $\frac{R_p}{\sqrt{2}}$.
- The Quality Factor (Q) of the resonant circuit is defined as:

$$Q = \frac{\omega_0 L}{R} = \frac{f_0}{BW} \implies BW = f_2 - f_1 = \frac{f_0}{Q} = \frac{1}{2\pi R_p C}$$

Final Verified Result

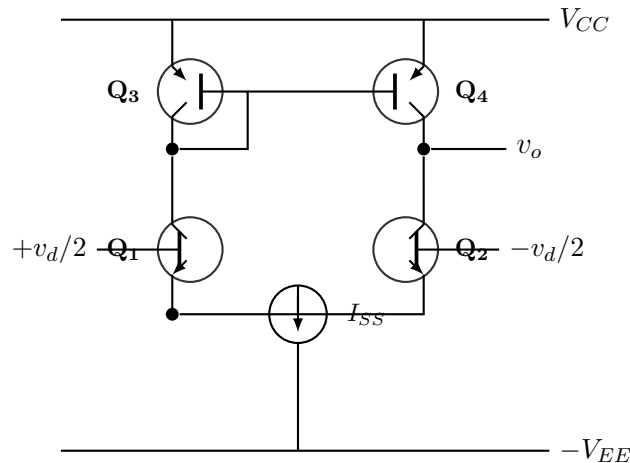
$$BW = \frac{f_0}{Q}$$

3.0.8 Question 8: Differential Amplifier with Active vs. Passive Load [5 Marks]

Problem Statement (2082 Baishakh, Q8)

Show that the voltage gain of the differential amplifier with active load is twice that with passive load.

Circuit Schematic / Analysis Diagram: Differential Amplifier with Active Current Mirror Load



1. **Differential Amplifier with Passive Resistive Load (R_C):** For a single-ended output taken from one collector:

$$A_{d(\text{passive})} = \frac{1}{2} g_m R_C$$

2. **Differential Amplifier with Active Current Mirror Load (Q_3, Q_4):**

- When a differential input $v_d = v_1 - v_2$ is applied, transistor Q_1 current increases by $i_c = \frac{1}{2} g_m v_d$ and transistor Q_2 current decreases by $i_c = -\frac{1}{2} g_m v_d$.
- The current mirror replicates the current change of Q_1 to the collector of Q_4 : $i_{c4} = i_{c1} = +\frac{1}{2} g_m v_d$.
- At the single-ended output node, the currents add constructively:

$$i_{\text{out}} = i_{c4} - i_{c2} = \left(+\frac{1}{2} g_m v_d \right) - \left(-\frac{1}{2} g_m v_d \right) = \mathbf{g_m v_d}$$

- The total output resistance is $R_{\text{out}} = r_{o2} \parallel r_{o4}$.

$$v_o = i_{\text{out}} R_{\text{out}} = (g_m v_d)(r_{o2} \parallel r_{o4}) \implies \mathbf{A_{d(active)} = g_m(r_{o2} \parallel r_{o4})}$$

Thus, the active current mirror provides differential-to-single-ended conversion that doubles the effective transconductance, making the active load gain **twice** the single-ended passive load gain ($A_{d(\text{active})} = 2 \times A_{d(\text{passive})}$). ■

Final Verified Result

$$\mathbf{A_{d(active)} = g_m R_{\text{out}} = 2 \times \left(\frac{1}{2} g_m R_C \right) = 2 \times A_{d(\text{passive})}}$$

3.0.9 Question 9: Efficiency of Series-Fed Class A Amplifier [4 Marks]

Problem Statement (2082 Baishakh, Q9)

Derive general efficiency of series fed class A amplifier.

In a series-fed direct-coupled Class A amplifier:

1. **DC Input Power:** $P_{i(dc)} = V_{CC}I_{CQ}$.
2. **AC Output Power:** $P_{o(ac)} = \frac{V_p I_p}{2} = \frac{V_p^2}{2R_C}$.
3. **Conversion Efficiency:**

$$\eta = \frac{P_{o(ac)}}{P_{i(dc)}} = \frac{\frac{V_p I_p}{2}}{V_{CC}I_{CQ}} = \frac{1}{2} \left(\frac{V_p}{V_{CC}} \right) \left(\frac{I_p}{I_{CQ}} \right) \times 100\%$$

4. **Maximum Theoretical Efficiency:** Since $V_p \leq \frac{V_{CC}}{2}$ and $I_p \leq I_{CQ}$ in a resistive series-fed collector circuit:

$$\eta_{\max} = \frac{1}{2} \left(\frac{1}{2} \right) (1) \times 100\% = 25\%$$

Final Verified Result

$$\eta = \frac{1}{2} \left(\frac{V_p}{V_{CC}} \right) \left(\frac{I_p}{I_{CQ}} \right) \times 100\%, \quad \eta_{\max} = 25\%$$

3.0.10 Question 10: Class B Maximum Power Capabilities [5 Marks]

Problem Statement (2082 Baishakh, Q10)

For a class B amplifier using a supply of $V_{CC} = 30V$ and driving a load of 16Ω , determine the maximum input power, maximum output power and maximum power dissipation at transistors.

1. **Maximum AC Output Power ($P_{o(max)}$):** Occurs at maximum linear voltage swing $V_p = V_{CC} = 30V$:

$$P_{o(max)} = \frac{V_{CC}^2}{2R_L} = \frac{(30V)^2}{2 \times 16 \Omega} = \frac{900}{32} = 28.125 \text{ W}$$

2. **Maximum DC Input Power ($P_{i(max)}$):**

$$I_{dc(max)} = \frac{2}{\pi} \left(\frac{V_{CC}}{R_L} \right) = \frac{2}{\pi} \times \frac{30V}{16 \Omega} = \frac{60}{16\pi} \approx 1.1937 \text{ A}$$

$$P_{i(max)} = V_{CC} \times I_{dc(max)} = \frac{2V_{CC}^2}{\pi R_L} = \frac{2(900)}{16\pi} = 35.81 \text{ W}$$

3. **Maximum Transistor Power Dissipation ($P_{D(max)}$):** Transistor power dissipation is $P_D = P_i -$

$$P_o = \frac{2V_{CC}V_p}{\pi R_L} - \frac{V_p^2}{2R_L}.$$

Differentiating with respect to V_p and setting to zero gives the worst-case heating condition at $V_p = \frac{2}{\pi}V_{CC} \approx 0.6366V_{CC}$:

$$P_{D(\max)} = \frac{2V_{CC}^2}{\pi^2 R_L} = \frac{2 \times (30V)^2}{\pi^2 \times 16 \Omega} = \frac{1800}{157.91} = 11.399 \text{ W}$$

Per transistor dissipation: $P_{D1(\max)} = \frac{P_{D(\max)}}{2} = 5.70 \text{ W}.$

Final Verified Result

$$P_{O(\max)} = 28.125 \text{ W}, \quad P_{i(\max)} = 35.81 \text{ W}, \quad P_{D(\max)} = 11.40 \text{ W}$$

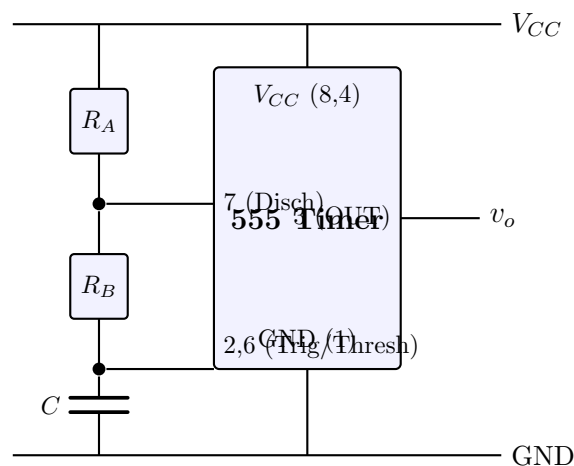
3.0.11 Question 11: 555 Timer Astable Multivibrator Frequency Derivation [4 Marks]

Problem Statement (2082 Baishakh, Q11)

Derive frequency of oscillation of 555 timer Astable Multivibrator.

In an Astable 555 multivibrator, capacitor C charges through $(R_A + R_B)$ from $\frac{1}{3}V_{CC}$ to $\frac{2}{3}V_{CC}$ and discharges through R_B from $\frac{2}{3}V_{CC}$ to $\frac{1}{3}V_{CC}$.

Circuit Schematic / Analysis Diagram: 555 Timer Astable Multivibrator Circuit



1. Charging Time (T_{HIGH}):

$$v_C(t) = V_{CC} - \frac{2}{3}V_{CC}e^{-t/(R_A+R_B)C} = \frac{2}{3}V_{CC} \implies T_{\text{HIGH}} = (R_A + R_B)C \ln(2) \approx 0.693(R_A + R_B)C$$

2. Discharging Time (T_{LOW}):

$$v_C(t) = \frac{2}{3}V_{CC}e^{-t/R_B C} = \frac{1}{3}V_{CC} \Rightarrow T_{\text{LOW}} = R_B C \ln(2) \approx 0.693R_B C$$

3. Total Period and Frequency:

$$T = T_{\text{HIGH}} + T_{\text{LOW}} = 0.693(R_A + 2R_B)C \Rightarrow f = \frac{1}{T} = \frac{1.44}{(R_A + 2R_B)C}$$

Final Verified Result

$$f = \frac{1.44}{(R_A + 2R_B)C}, \quad \text{Duty Cycle } D = \frac{R_A + R_B}{R_A + 2R_B} \times 100\%$$

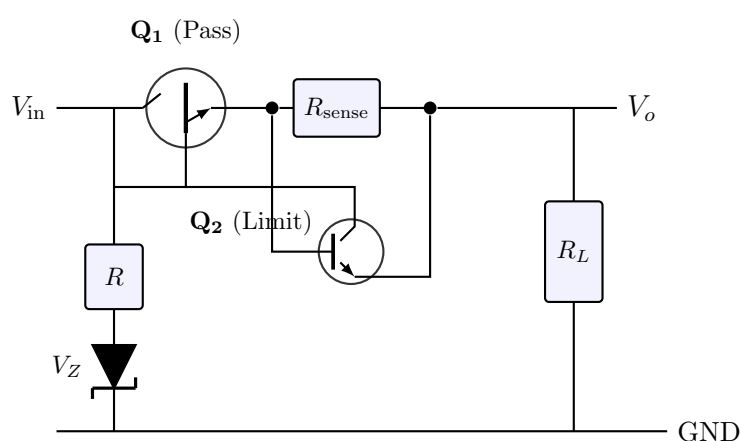
3.0.12 Question 12: Series Voltage Regulator with Current Limiting & LM317 Design [4 + 4 = 8 Marks]

Problem Statement (2082 Baishakh, Q12)

Draw series voltage regulator with current limiting circuit and explain how this protection circuit works. Design a voltage regulator to give output voltage from 5V to 15V using LM317.

Part (a): Current Limiting Protection Circuit [4 Marks]

Circuit Schematic / Analysis Diagram: Series Transistor Voltage Regulator with Current Limiting Protection



Circuit Operation: A current-sense resistor R_{sense} and protection transistor Q_2 are placed in series with pass transistor Q_1 :

- **Normal Operation** ($I_L < I_{\text{limit}}$): The voltage drop across R_{sense} is less than the cut-in voltage

($I_L R_{\text{sense}} < 0.6\text{V}$). Transistor Q_2 is OFF and does not interfere with regular voltage regulation.

- **Overload / Short-Circuit Condition ($I_L \geq I_{\text{limit}}$):** When $I_L R_{\text{sense}} \geq 0.7\text{V}$, transistor Q_2 turns ON, diverting excess base drive current away from pass transistor Q_1 . This limits the maximum output current to a safe threshold:

$$I_{\text{limit}} = \frac{V_{\text{BE2}}}{R_{\text{sense}}} \approx \frac{0.7\text{V}}{R_{\text{sense}}}$$

Part (b): LM317 Design for 5V – 15V Output [4 Marks]

$$V_{\text{out}} = 1.25\text{V} \left(1 + \frac{R_2}{R_1} \right)$$

1. Select standard $R_1 = 240 \, \Omega$.

2. For $V_{\text{out}(\text{min})} = 5.0\text{V}$:

$$5.0 = 1.25 \left(1 + \frac{R_{2(\text{min})}}{240} \right) \implies 1 + \frac{R_{2(\text{min})}}{240} = 4.0 \implies R_{2(\text{min})} = 3 \times 240 = 720 \, \Omega$$

3. For $V_{\text{out}(\text{max})} = 15.0\text{V}$:

$$15.0 = 1.25 \left(1 + \frac{R_{2(\text{max})}}{240} \right) \implies 1 + \frac{R_{2(\text{max})}}{240} = 12.0 \implies R_{2(\text{max})} = 11 \times 240 = 2640 \, \Omega = 2.64 \, \text{k}\Omega$$

Implementation: Use a fixed resistor $R_{2A} = 680 \, \Omega$ in series with a **2.0 k Ω** linear potentiometer.

Final Verified Result

$$R_1 = 240 \, \Omega, \quad R_2 = 720 \, \Omega \text{ to } 2.64 \, \text{k}\Omega \quad (\text{Fixed } 680 \, \Omega + 2.0\text{k}\Omega \text{ Potentiometer})$$

Chapter 4

2081 Ashwin Examination Solutions

Examination Overview

Level: Bachelor in Engineering (BE)

Full Marks: 60

Programme: BEI, BCT

Pass Marks: 24

Year / Part: I / II

Time: 3 Hours

Subject: Electronic Devices & Circuits (EX 151)

Examination Type: Regular (New Course - 2080 Batch)

4.0.1 Question 1: Stiff β -Independent CE Amplifier Design [4 Marks]

Problem Statement (2081 Ashwin, Q1)

Design β -independent type DC biased common emitter amplifier. Given parameters: $V_{CC} = 20\text{V}$, $I_C = 1.5\text{mA}$ and $\beta = 110$. Use stiff biasing method.

1. **Emitter Voltage Allocation (V_E):** Set $V_E = 0.10 \times V_{CC} = 0.10 \times 20\text{V} = \mathbf{2.0\text{V}}$.

$$R_E = \frac{V_E}{I_E} \approx \frac{2.0\text{V}}{1.5\text{mA}} = 1.33 \text{ k}\Omega \quad (\text{Select standard } R_E = \mathbf{1.3 \text{ k}\Omega} \text{ or } \mathbf{1.5 \text{ k}\Omega})$$

2. **Collector Resistor Allocation (R_C):** Set $V_{CEQ} = \frac{V_{CC} - V_E}{2} = \frac{20 - 2.0}{2} = 9.0\text{V}$:

$$R_C = \frac{V_{CC} - (V_E + V_{CEQ})}{I_C} = \frac{20 - 11.0}{1.5\text{mA}} = \frac{9.0\text{V}}{1.5\text{mA}} = \mathbf{6.0 \text{ k}\Omega} \quad (\text{Select standard } R_C = \mathbf{6.2 \text{ k}\Omega})$$

3. **Base Voltage Calculation (V_B):**

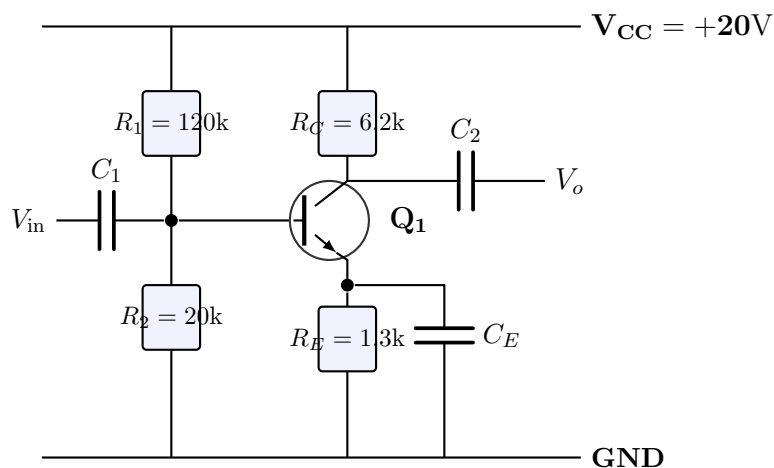
$$V_B = V_E + V_{BE} = 2.0\text{V} + 0.7\text{V} = \mathbf{2.7 \text{ V}}$$

4. Stiff Biasing Bleeder Current Rule ($I_{\text{bleed}} \geq 10I_B$):

$$I_B = \frac{1.5\text{mA}}{110} = 13.64 \mu\text{A} \Rightarrow I_{\text{bleed}} = 10 \times 13.64 \mu\text{A} = 136.4 \mu\text{A} = 0.1364\text{mA}$$

$$R_2 = \frac{V_B}{I_{\text{bleed}}} = \frac{2.7\text{V}}{0.1364\text{mA}} = 19.79 \text{ k}\Omega \quad (\text{Select standard } R_2 = \mathbf{20 \text{ k}\Omega} \text{ or } \mathbf{18 \text{ k}\Omega})$$

$$R_1 = \frac{V_{CC} - V_B}{I_{\text{bleed}} + I_B} = \frac{20\text{V} - 2.7\text{V}}{0.15\text{mA}} = \frac{17.3\text{V}}{0.15\text{mA}} = 115.33 \text{ k}\Omega \quad (\text{Select standard } R_1 = \mathbf{120 \text{ k}\Omega})$$

Circuit Schematic / Analysis Diagram: Designed Stiff CE Amplifier Circuit ($V_{CC} = 20\text{V}$)

Final Verified Result

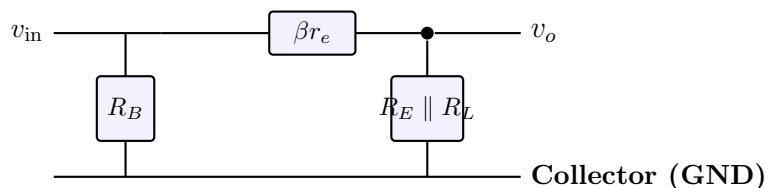
$$R_1 = 120 \text{ k}\Omega, \quad R_2 = 20 \text{ k}\Omega, \quad R_C = 6.2 \text{ k}\Omega, \quad R_E = 1.3 \text{ k}\Omega$$

4.0.2 Question 2: Common Collector (Emitter Follower) Small Signal Analysis [1 + 4 = 5 Marks]

Problem Statement (2081 Ashwin, Q2)

Why is a common collector BJT amplifier also known as an emitter follower? Draw the small signal model of common collector amplifier circuit and derive expressions for input resistance, voltage gain, and current gain.

Part (a): Why Termed "Emitter Follower" [1 Mark] In a Common Collector configuration, the output voltage is extracted at the emitter terminal. Since $v_o = v_e = v_{in} - v_{be} \approx v_{in}$ and the voltage gain is nearly unity ($A_v \approx 0.99 \approx 1$) with zero phase shift (0°), the emitter AC output potential **closely tracks and follows the input signal**, hence the name **Emitter Follower**.

Part (b): Parameter Derivations [4 Marks]**Circuit Schematic / Analysis Diagram: Small-Signal Model of Common Collector (Emitter Follower)**

1. **Input Resistance (R_{in}):** Looking into the base terminal:

$$Z_b = \beta r_e + (1 + \beta)(R_E \parallel R_L) \approx \beta(r_e + R_E \parallel R_L) \quad (\text{very high, } > 100 \text{ k}\Omega)$$

Total amplifier input resistance: $R_{in} = R_1 \parallel R_2 \parallel Z_b$.

2. **Voltage Gain (A_v):**

$$v_o = i_e(R_E \parallel R_L), \quad v_{in} = i_e[r_e + (R_E \parallel R_L)]$$

$$A_v = \frac{v_o}{v_{in}} = \frac{R_E \parallel R_L}{r_e + R_E \parallel R_L} \approx 1 \quad (\text{Non-inverting, } 0^\circ \text{ phase shift})$$

3. **Current Gain (A_i):**

$$A_i = \frac{i_o}{i_{in}} = (1 + \beta) \left(\frac{R_B}{R_B + Z_b} \right) \left(\frac{R_E}{R_E + R_L} \right) \approx \beta \quad (\text{very high})$$

Final Verified Result

$$R_{in} \approx R_B \parallel \beta R_E, \quad A_v = \frac{R_E \parallel R_L}{r_e + R_E \parallel R_L} \approx 1, \quad A_i \approx \beta$$

4.0.3 Question 3: Derivation of BJT Transconductance g_m [4 Marks]**Problem Statement (2081 Ashwin, Q3)**

Derive trans-conductance of BJT.

The transconductance g_m quantifies the small-signal sensitivity of the collector output current with respect to changes in the base-emitter input voltage:

$$g_m = \left. \frac{\partial i_C}{\partial v_{BE}} \right|_{Q\text{-point}}$$

Derivation from Shockley Diode Conduction Equation: The collector current of a BJT operating in the active region is given by:

$$i_C = I_S e^{v_{BE}/V_T}$$

where I_S is the reverse saturation current and $V_T = \frac{kT}{q} \approx 26 \text{ mV}$ is the thermal voltage at room temperature (300 K).

Differentiating with respect to v_{BE} :

$$\frac{di_C}{dv_{BE}} = I_S \cdot \frac{d}{dv_{BE}} \left(e^{v_{BE}/V_T} \right) = I_S \cdot \left(\frac{1}{V_T} e^{v_{BE}/V_T} \right) = \frac{I_S e^{v_{BE}/V_T}}{V_T}$$

Recognizing that $I_S e^{v_{BEQ}/V_T} = I_C$ (quiescent collector current):

$$g_m = \frac{I_C}{V_T} = \frac{1}{r_e} \quad \blacksquare$$

Final Verified Result

$$g_m = \frac{I_C}{V_T} \approx \frac{I_C}{26 \text{ mV}}$$

4.0.4 Question 4: N-Channel Enhancement-Type MOSFET [7 Marks]

Problem Statement (2081 Ashwin, Q4)

Describe the construction and working principle of n channel enhancement-type MOSFET with the help of necessary diagrams and its drain characteristic curve.

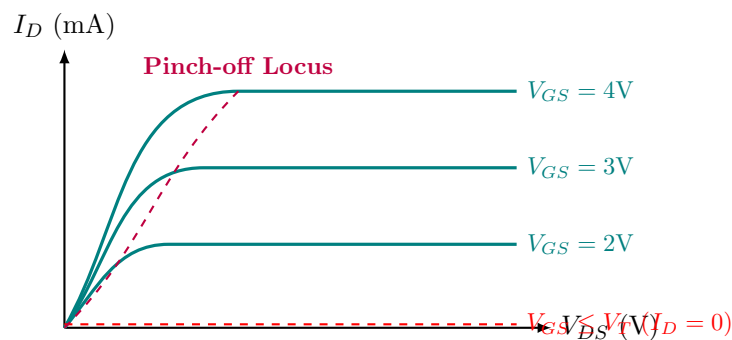
1. Physical Construction: An N-Channel Enhancement MOSFET (E-MOSFET) is fabricated on a p-type substrate without any pre-existing channel:

- Two heavily doped n^+ regions form the Source and Drain.
- A thin dielectric insulating layer of SiO_2 separates the metal/polysilicon Gate electrode from the substrate.
- With $V_{GS} = 0\text{V}$, two back-to-back p - n junctions block all current flow between drain and source ($I_D = 0$).

2. Working Principle and Inversion Layer Formation:

- When a positive gate voltage is applied ($V_{GS} > 0$), the electric field repels holes away from the surface and attracts minority electrons.
- When $V_{GS} \geq V_T$ (**Threshold Voltage**, typically +1V to +3V), an **inversion layer of electrons** forms beneath the gate oxide, creating an induced n-channel bridging source and drain.

Circuit Schematic / Analysis Diagram: Drain Characteristic Curves of N-Channel E-MOSFET



3. Drain Current Conduction Equations:

- Ohmic / Triode Region ($V_{DS} < V_{GS} - V_T$):

$$I_D = K [2(V_{GS} - V_T)V_{DS} - V_{DS}^2]$$

- Saturation Region ($V_{DS} \geq V_{GS} - V_T$):

$$I_D = K(V_{GS} - V_T)^2 = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right) (V_{GS} - V_T)^2$$

Final Verified Result

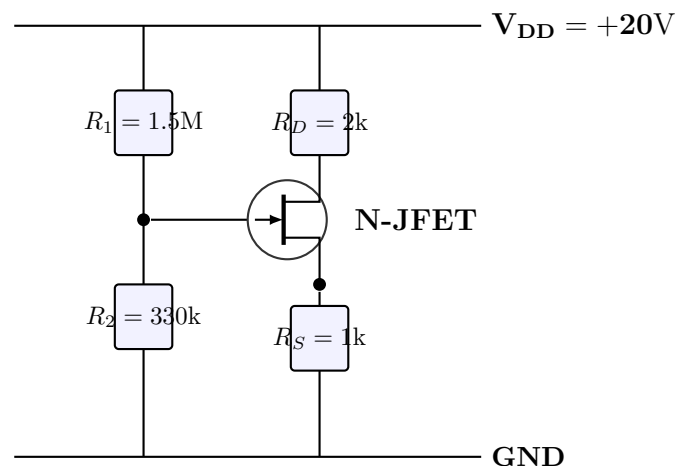
$$I_D = K(V_{GS} - V_T)^2 \quad \text{for } V_{GS} \geq V_T \text{ and } V_{DS} \geq V_{GS} - V_T$$

4.0.5 Question 5: JFET Q-Point and Pinch-Off Region Verification [6 Marks]

Problem Statement (2081 Ashwin, Q5)

Find I_D and V_{DS} for the following circuit. Given data are $V_P = -5.5V$, $I_{DSS} = 10mA$. Assume all capacitors are ideal and check whether the transistor is operating in the pinch-off region or not. Circuit parameters: $V_{DD} = 20V$, $R_1 = 1.5 \text{ M}\Omega$, $R_2 = 330 \text{ k}\Omega$, $R_D = 2.0 \text{ k}\Omega$, $R_S = 1.0 \text{ k}\Omega$.

Circuit Schematic / Analysis Diagram: Given JFET Circuit Schematic

1. Gate Voltage Calculation (V_G):

$$V_G = V_{DD} \left(\frac{R_2}{R_1 + R_2} \right) = 20\text{V} \times \left(\frac{330\text{k}\Omega}{1500\text{k}\Omega + 330\text{k}\Omega} \right) = 20 \times \frac{0.33}{1.83} = \mathbf{3.6066\text{ V}}$$

2. Equation for V_{GS} :

$$V_{GS} = V_G - I_D R_S = 3.6066 - 1.0 I_D \quad (\text{with } I_D \text{ in mA})$$

3. Substitution into Shockley's Equation:

$$I_D = 10 \left(1 - \frac{3.6066 - 1.0 I_D}{-5.5} \right)^2 = 10 \left(\frac{5.5 + 3.6066 - I_D}{5.5} \right)^2 = \frac{10}{30.25} (9.1066 - I_D)^2$$

$$3.025 I_D = 82.93 - 18.213 I_D + I_D^2 \implies I_D^2 - 21.238 I_D + 82.93 = 0$$

Solving via quadratic formula:

$$I_D = \frac{21.238 \pm \sqrt{(21.238)^2 - 4(1)(82.93)}}{2} = \frac{21.238 \pm \sqrt{451.05 - 331.72}}{2} = \frac{21.238 \pm 10.924}{2}$$

Two mathematical roots:

- $I_{D1} = 16.081\text{ mA} \implies V_{GS1} = 3.6066 - 16.081 = -12.47\text{V} < V_P$ (**Invalid**).
- $I_{D2} = \mathbf{5.157\text{ mA}} \implies V_{GS2} = 3.6066 - 5.157 = \mathbf{-1.550\text{ V}}$ (**Valid!**).

4. Calculation of Drain-to-Source Voltage (V_{DS}):

$$V_{DS} = V_{DD} - I_D (R_D + R_S) = 20\text{V} - 5.157\text{mA} \times (2.0\text{k}\Omega + 1.0\text{k}\Omega) = 20 - 5.157(3.0) = \mathbf{4.529\text{ V}}$$

5. Verification of Pinch-Off Region: The minimum drain-to-source voltage required for pinch-off (saturation) is:

$$V_{DS(\text{sat})} = V_{GS} - V_P = -1.550\text{V} - (-5.5\text{V}) = \mathbf{3.950\text{ V}}$$

Since $V_{DS} = 4.529\text{V} \geq V_{DS(\text{sat})} = 3.950\text{V}$, the **JFET is definitively operating in the pinch-off (saturation) region.**

Final Verified Result

$$\mathbf{I_D = 5.157\text{ mA}, \quad V_{GS} = -1.550\text{ V}, \quad V_{DS} = 4.529\text{ V}, \quad V_{DS} > (V_{GS} - V_P) \Rightarrow \text{Confirmed in Pinch-off Region}}$$

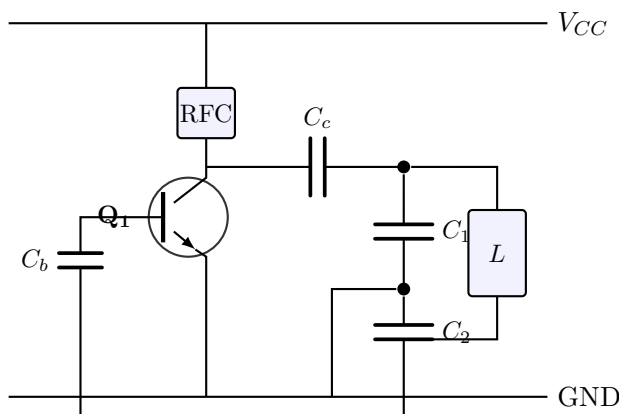
4.0.6 Question 6: Colpitts Oscillator Schematic & Frequency Derivation [4 Marks]

Problem Statement (2081 Ashwin, Q6)

State Barkhausen criterion for sinusoidal oscillation. Draw the circuit diagram of Colpitts oscillator and write its frequency of oscillation.

A **Colpitts Oscillator** utilizes a capacitive voltage divider composed of two capacitors (C_1, C_2) in parallel with an inductor L in its feedback tank.

Circuit Schematic / Analysis Diagram: BJT Colpitts Oscillator Circuit



Frequency of Oscillation: The series combination of C_1 and C_2 gives the equivalent tank capacitance:

$$C_{\text{eq}} = \frac{C_1 C_2}{C_1 + C_2} \Rightarrow f_0 = \frac{1}{2\pi \sqrt{L C_{\text{eq}}}} = \frac{1}{2\pi \sqrt{L \left(\frac{C_1 C_2}{C_1 + C_2} \right)}}$$

Final Verified Result

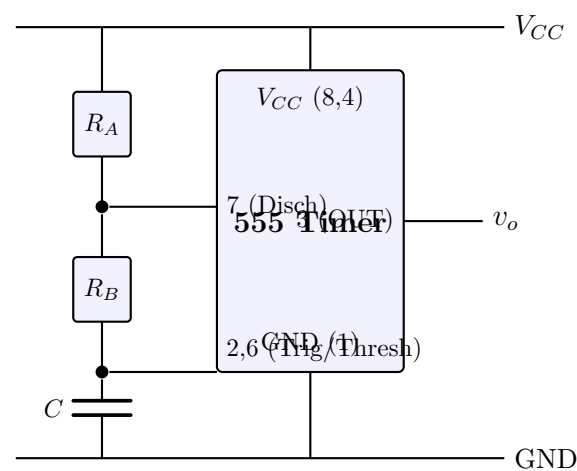
$$f_0 = \frac{1}{2\pi\sqrt{LC_{eq}}}, \quad C_{eq} = \frac{C_1 C_2}{C_1 + C_2}$$

4.0.7 Question 7: 555 Timer Astable Multivibrator Frequency Derivation [4 Marks]

Problem Statement (2081 Ashwin, Q7)

Derive frequency of oscillation of 555 timer Astable Multivibrator.

Circuit Schematic / Analysis Diagram: 555 Timer Astable Multivibrator Circuit

1. Charging Time (T_{HIGH}):

$$v_C(t) = V_{CC} - \frac{2}{3}V_{CC}e^{-t/(R_A+R_B)C} = \frac{2}{3}V_{CC} \Rightarrow T_{HIGH} = (R_A + R_B)C \ln(2) \approx 0.693(R_A + R_B)C$$

2. Discharging Time (T_{LOW}):

$$v_C(t) = \frac{2}{3}V_{CC}e^{-t/R_B C} = \frac{1}{3}V_{CC} \Rightarrow T_{LOW} = R_B C \ln(2) \approx 0.693R_B C$$

3. Total Period and Frequency:

$$T = T_{HIGH} + T_{LOW} = 0.693(R_A + 2R_B)C \Rightarrow f = \frac{1}{T} = \frac{1.44}{(R_A + 2R_B)C}$$

Final Verified Result

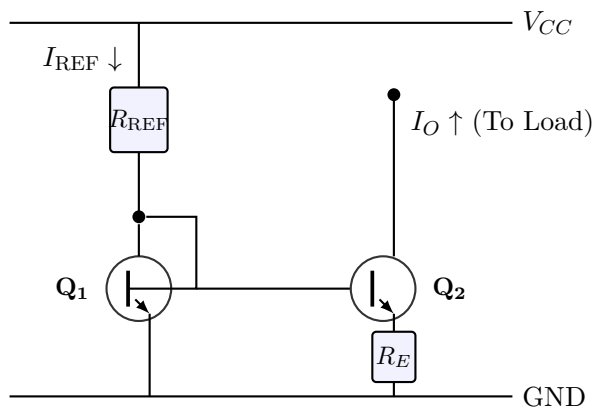
$$f = \frac{1.44}{(R_A + 2R_B)C}, \quad T_{HIGH} = 0.693(R_A + R_B)C, \quad T_{LOW} = 0.693R_B C$$

4.0.8 Question 8: Widlar Current Source Output Resistance Derivation [5 Marks]

Problem Statement (2081 Ashwin, Q8)

Draw a Widlar Current Source. Derive an expression for an output resistance of Widlar Current Source.

Circuit Schematic / Analysis Diagram: Widlar Current Source Circuit Diagram



Applying small-signal analysis to the output transistor Q_2 with emitter resistance R_E :

1. Looking into the collector of Q_2 , the effective emitter-to-ground impedance is $R'_E = R_E \parallel r_{\pi 2}$.
2. Taking the early-effect output resistance $r_{o2} = \frac{V_A}{I_O}$ into account, the small-signal KVL at the collector gives:

$$i_o = \frac{v_o - v_e}{r_{o2}} + g_{m2}v_{be2}$$

where $v_e = i_o(R_E \parallel r_{\pi 2})$ and $v_{be2} = -v_e = -i_o(R_E \parallel r_{\pi 2})$.

$$i_o = \frac{v_o - i_o(R_E \parallel r_{\pi 2})}{r_{o2}} - g_{m2}i_o(R_E \parallel r_{\pi 2})$$

$$v_o = i_o [r_{o2} + (R_E \parallel r_{\pi 2}) + g_{m2}r_{o2}(R_E \parallel r_{\pi 2})]$$

$$\mathbf{R}_{\text{out}} = \frac{v_o}{i_o} = \mathbf{r}_{o2} [1 + \mathbf{g}_{m2}(\mathbf{R}_E \parallel \mathbf{r}_{\pi 2})] + (\mathbf{R}_E \parallel \mathbf{r}_{\pi 2}) \approx \mathbf{r}_{o2} (1 + \mathbf{g}_{m2}\mathbf{R}_E)$$

Significance: The emitter resistor R_E provides negative current feedback that multiplies the intrinsic output resistance r_{o2} by a factor of $(1 + g_{m2}R_E)$, providing an exceptionally stable, high-impedance current source.

Final Verified Result

$$\mathbf{R}_{\text{out}} \approx \mathbf{r}_{o2} [1 + \mathbf{g}_{m2}(\mathbf{R}_E \parallel \mathbf{r}_{\pi 2})] \approx \mathbf{r}_{o2}(1 + \mathbf{g}_{m2}\mathbf{R}_E)$$

4.0.9 Question 9: Efficiency Calculation of Transformer-Coupled Class A Amplifier [4 Marks]

Problem Statement (2081 Ashwin, Q9)

Calculate the efficiency of a transformer coupled Class A amplifier for a supply of 12V DC and output of $V_p = 12V$ and 6V.

For an ideal transformer-coupled Class A amplifier:

$$\eta = \frac{1}{2} \left(\frac{V_p}{V_{CC}} \right)^2 \times 100\%$$

1. **Case 1: For $V_p = 12.0V$ (Maximum Linear Swing):**

$$\eta = \frac{1}{2} \left(\frac{12V}{12V} \right)^2 \times 100\% = \frac{1}{2} (1.0)^2 \times 100\% = \mathbf{50.0\%}$$

2. **Case 2: For $V_p = 6.0V$ (Half Voltage Swing):**

$$\eta = \frac{1}{2} \left(\frac{6V}{12V} \right)^2 \times 100\% = \frac{1}{2} (0.5)^2 \times 100\% = \frac{1}{2} (0.25) \times 100\% = \mathbf{12.5\%}$$

Final Verified Result

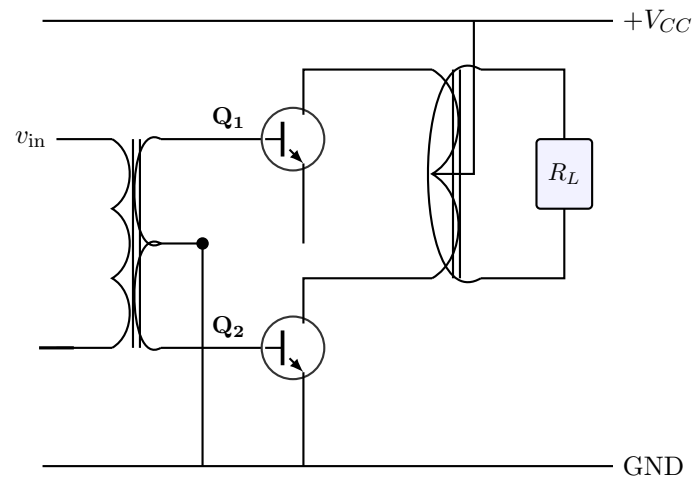
$$\text{At } V_p = 12V : \quad \eta = 50.0\%, \quad \text{At } V_p = 6V : \quad \eta = 12.5\%$$

4.0.10 Question 10: Transformer-Coupled Class B Push-Pull Amplifier Efficiency [5 Marks]

Problem Statement (2081 Ashwin, Q10)

Draw the circuit diagram and the characteristic curve of a transformer coupled class B push-pull amplifier and derive its general efficiency.

Circuit Schematic / Analysis Diagram: Transformer-Coupled Class B Push-Pull Amplifier Circuit



Analytical Derivation of General Efficiency:

1. AC Output Power (P_o):

$$P_{o(ac)} = \frac{V_p^2}{2R'_L}$$

2. DC Input Power (P_i): The average current drawn from supply V_{CC} across both half cycles is:

$$I_{dc} = \frac{2}{\pi} I_p = \frac{2}{\pi} \left(\frac{V_p}{R'_L} \right) \Rightarrow P_{i(dc)} = V_{CC} I_{dc} = \frac{2}{\pi} \frac{V_{CC} V_p}{R'_L}$$

3. Conversion Efficiency (η):

$$\eta = \frac{P_{o(ac)}}{P_{i(dc)}} = \frac{\frac{V_p^2}{2R'_L}}{\frac{2V_{CC}V_p}{\pi R'_L}} = \frac{\pi}{4} \left(\frac{V_p}{V_{CC}} \right) \times 100\%$$

At maximum linear swing ($V_p = V_{CC}$): $\eta_{\max} = \frac{\pi}{4} \approx 78.54\%$.

Final Verified Result

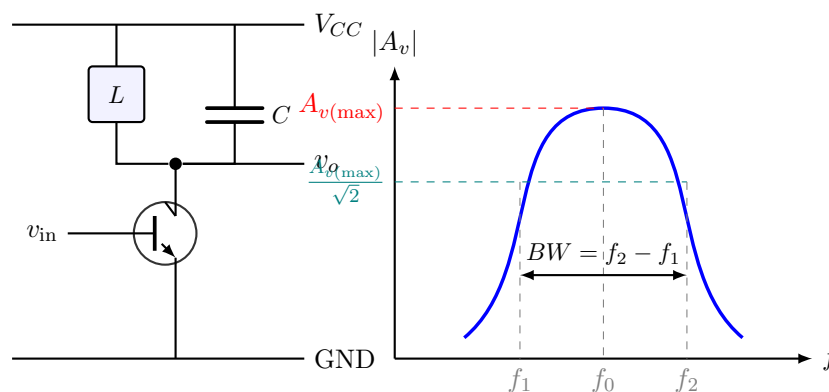
$$\eta = \frac{\pi}{4} \left(\frac{V_p}{V_{CC}} \right) \times 100\%, \quad \eta_{\max} = 78.54\%$$

4.0.11 Question 11: Class A Tuned Amplifier 3dB Bandwidth [4 Marks]

Problem Statement (2081 Ashwin, Q11)

Draw the circuit diagram of class A tuned amplifier with its frequency response. Derive its 3dB bandwidth.

Circuit Schematic / Analysis Diagram: Class A Tuned Amplifier Circuit and Resonant Response



Derivation of 3dB Bandwidth: The resonant frequency of the parallel LC tank is $f_0 = \frac{1}{2\pi\sqrt{LC}}$. The loaded quality factor Q is defined as $Q = \frac{R_p}{\omega_0 L} = \omega_0 C R_p$.

The 3dB bandwidth between lower (f_1) and upper (f_2) half-power cutoff frequencies is:

$$BW = f_2 - f_1 = \frac{f_0}{Q} = \frac{1}{2\pi R_p C}$$

Final Verified Result

$$BW = \frac{f_0}{Q} = \frac{1}{2\pi R_p C}$$

4.0.12 Question 12: LM317 Variable DC Regulator Design (10V - 20V) [4 Marks]

Problem Statement (2081 Ashwin, Q12)

Design variable DC voltage regulator using LM 317 to get (10 – 20) volts output.

$$V_{out} = 1.25V \left(1 + \frac{R_2}{R_1} \right)$$

1. Select standard $R_1 = 240 \, \Omega$.
2. For $V_{\text{out}(\min)} = 10.0\text{V}$:

$$10.0 = 1.25 \left(1 + \frac{R_{2(\min)}}{240} \right) \Rightarrow R_{2(\min)} = 7.0 \times 240 = 1680 \, \Omega = 1.68 \, \text{k}\Omega$$

3. For $V_{\text{out}(\max)} = 20.0\text{V}$:

$$20.0 = 1.25 \left(1 + \frac{R_{2(\max)}}{240} \right) \Rightarrow R_{2(\max)} = 15.0 \times 240 = 3600 \, \Omega = 3.60 \, \text{k}\Omega$$

Implementation: Use a fixed resistor $R_{2A} = 1.6 \, \text{k}\Omega$ in series with a **2.0 k Ω** linear potentiometer ($R_2 = 1.68 \, \text{k}\Omega$ to $3.60 \, \text{k}\Omega$). Input voltage: $V_{\text{in}} \geq 23\text{V DC}$.

Final Verified Result

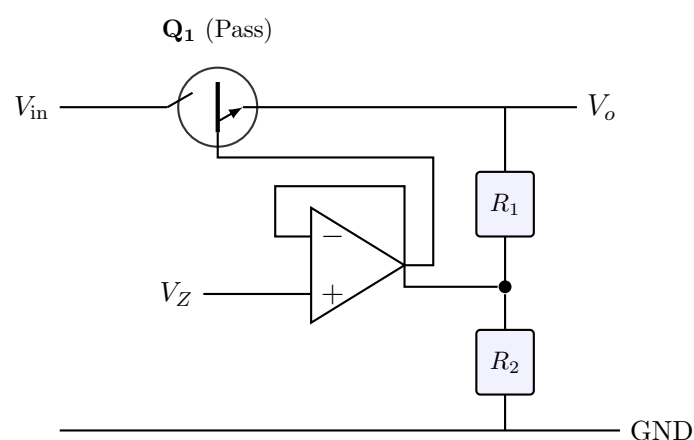
$R_1 = 240 \, \Omega$, $R_2 = 1.68 \, \text{k}\Omega$ to $3.60 \, \text{k}\Omega$ (Fixed $1.6\text{k}\Omega + 2.0\text{k}\Omega$ Potentiometer)

4.0.13 Question 13: Series Transistor Regulator with Error Amplifier & Stability Factor [2 + 2 = 4 Marks]

Problem Statement (2081 Ashwin, Q13)

Explain the working principle of a series transistor voltage regulator with improved performance and derive an expression for the voltage stability.

Circuit Schematic / Analysis Diagram: Series Voltage Regulator with Op-Amp / BJT Error Amplifier



An improved series regulator incorporates an active error amplifier (Op-Amp or BJT Q_2) comparing a sampled fraction of the output $\beta_s V_o = \frac{R_2}{R_1 + R_2} V_o$ against reference voltage V_Z .

- **Operation:** If V_o increases due to input or load variations, V_{sample} increases above V_Z , causing the error amplifier output to pull down the base potential of pass transistor Q_1 , decreasing V_o back to the set level.
- **Voltage Stability Factor (S_V):**

$$S_V = \frac{\Delta V_o}{\Delta V_{\text{in}}} = \frac{r_z}{R + r_z} \cdot \frac{1}{1 + A_{\text{error}}\beta_s} \approx \frac{r_z}{R \cdot A_{\text{error}}\beta_s}$$

The high open-loop gain A_{error} of the error amplifier suppresses input line fluctuations by several orders of magnitude.

Final Verified Result

$$S_V = \frac{\Delta V_o}{\Delta V_{\text{in}}} \approx \frac{r_z}{R \cdot A_{\text{error}}\beta_s}$$